

# **Auditable Emergent Communication Between Isolated Artificial Agents**

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# A Preregistrable Study Protocol for Comparing Learning, Grounding, and Protocol Formation

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# Abstract

Research on emergent communication has shown that artificial agents can develop task-effective signaling protocols in referential games, multi-agent reinforcement learning, embodied environments, and negotiation. However, successful coordination does not by itself establish that a protocol is grounded, compositionally generalizable, causally used by a receiver, or interpretable. Comparisons are further complicated when pretrained language models, agents trained from scratch, different communication carriers, and different reward mechanisms are evaluated in separate experimental systems.

We present the design of the **Agentic Language Development Nursery Lab**, a preregistrable framework for studying communication between two isolated artificial agents through a deterministic, bandwidth-constrained gateway. The framework is designed to compare frozen language-model adaptation, from-scratch multi-agent reinforcement learning, intrinsic-motivation learning, self-supervised learning, hybrid models, and no-learning controls under common scenario, channel, evidence, and evaluation interfaces. Each agent is specified to maintain an independent chronological ledger of its own intentions and interpretations. The framework specifies append-only, hash-chained, signed ledger and channel records, ordered Merkle checkpoints, and deterministic simulated commitments. The local integrity, checkpoint, verifier, and commitment paths are implemented and tested under a zero-external-spend profile. Optional public-chain transport remains outside the approved campaign and its claims. These mechanisms do not prove that an agent's interpretation is truthful; they are designed to preserve what was recorded and make later alteration detectable. Causal message interventions, held-out generalization, partner replacement, leakage tests, and control-channel conditions are therefore required in addition to task success.

This manuscript is a study-protocol and framework contribution. It defines research questions, falsifiable hypotheses, experimental controls, analysis plans, integrity claims, and governance boundaries before empirical outcomes are available. We specifically reject the claim that prompting a pretrained model to "act like a baby" makes it language-naive. Instead, the framework operationalizes limited observation, tool-only action, private memory, controlled plasticity, and grounded interaction. Future revisions will report results only after the corresponding experiment protocols, evidence bundles, and verification reports are completed.

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## Keywords

Emergent communication; multi-agent systems; language games; referential games; multi-agent reinforcement learning; self-supervised learning; compositionality; causal listening; developmental robotics; AI auditability; reproducible machine learning; cryptographic provenance.

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# Research Integrity Notice

This manuscript is structured to follow the Ethical Tech CoLab's *AI-Powered Assistance in Formulating Research Questions* guidance [47]:

- core sources are opened rather than accepted from an AI summary;
- citations and claims are checked against opened source or metadata pages;
- AI-suggested connections are treated as exploratory until supported;
- the influence and limitations of AI assistance are disclosed;
- source verification state and journal credibility are recorded in Appendix A;
- a separate critical-review pass is required before publication; the current internal pass is recorded in [reports/research/research-critical-review.md](#) and does not count as independent review.

In this draft, source retrieval and the first verification pass were performed by an AI assistant using web, scholarly-index, and metadata tools, followed by separate AI critical-review passes. Human-author re-verification is still pending and is a submission blocker, not an implied completed step.

The manuscript also adopts the CoLab's institutional ethical framing that ethical AI is a lifecycle and governance property rather than a product feature [46]. The study therefore treats protocol isolation, evidence integrity, human intervention, publication claims, and negative-result retention as part of the method.

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# 1. Introduction

## 1.1 Problem

Communication between artificial agents is often evaluated by whether it improves a shared task. This is necessary but insufficient. A sender can produce signals that correlate with its private observation while a receiver ignores them; two agents can memorize a holistic code that fails on novel combinations; a pretrained model can map arbitrary tokens onto concepts it already knows; and an apparently transparent post-hoc explanation can fail to reflect the mechanism that caused an action [7, 10-15].

These problems are particularly acute when the research question is framed through a developmental metaphor. A pretrained language model already contains extensive linguistic and cultural information. Calling such a model a "one-year-old" changes its role-play behavior, not its training history. The resulting simplified speech or childlike persona would be evidence of human-text imitation rather than first-language acquisition.

The central methodological challenge is therefore not to make a model *perform infancy*. It is to create an environment in which:

1. two agents have controlled and separately documented starting capabilities;
2. their only practical communication route is a deterministic public-artifact channel;
3. observations contain no unplanned human-language labels;
4. learning mechanisms can be varied without replacing the entire experimental system;
5. intentions, interpretations, actions, and outcomes are recorded independently;
6. causal use, generalization, leakage, and partner transfer can be tested;
7. the evidentiary history cannot be silently rewritten after a run.

## 1.2 Proposed Framework

The Nursery Lab instantiates two agent twins, **Learner A** and **Learner B**, and a third supervisory twin, the **Nursery Controller/BabySitter**. The learner twins may be described as "Babies" in the public research metaphor, but their model-facing contracts use the neutral term *Learner* and explicitly prohibit child role-play.

The learners exchange only validated artifacts through a deterministic Symbol Gateway. Baseline artifacts are symbols from a randomly assigned inventory with no provided meaning. Alternate experiments use unfamiliar glyphs, bitmaps, quantized strokes, or quantized tones. A six-display affect channel is disabled by default and, when enabled, is constrained to fixed post-outcome windows to reduce its capacity as a covert second language.

Each learner records its own intention and interpretation history. These private ledgers are not shared between learners. An authorized researcher can compare them after or during a run without feeding that comparison back to either agent. Causal interventions then test whether ledger claims predict behavior.

## 1.3 Unit of Contribution

This submission is one artifact with one unit of contribution: **a preregistrable, evidence-preserving experimental framework for controlled comparison of emergent communication across agent and learning types.**

It is not:

- an empirical claim that a new language has already emerged;
- a claim that an LLM can be made language-naïve by prompting;
- a theory of human infant cognition;
- a production cryptographic protocol;
- a claim that a ledger proves an agent's self-report is true.

## 1.4 Intended Contributions

If implemented and executed as specified, the work contributes:

1. **A common comparison interface.** Frozen LLM, from-scratch RL, intrinsic RL, self-supervised, hybrid, and no-learning agents operate under shared scenario, carrier, evidence, and evaluation contracts.
2. **Independent semantic histories.** Each learner maintains a separate chronological record of intended and inferred meanings.
3. **Causal ledger validation.** Message ablation, substitution, scrambling, and replay test whether ledger hypotheses predict receiver behavior.
4. **Controlled carrier comparison.** Fixed symbols and invented visual or acoustic forms can be compared without changing the broader experiment.
5. **Affect as an audited variable.** A low-capacity feedback channel is tested both for coordination value and covert information leakage.
6. **Evidence-first reproducibility.** Pre-registration, signed append-only records, Merkle consistency proofs, public checkpoint anchoring, and independent verification bind claims to a run history.
7. **A continuous cooperation-to-negotiation lineage.** The same established protocol can be observed as agent incentives move from aligned to partially conflicting.

## 2. Research Questions and Hypotheses

### 2.1 Primary Research Questions

**RQ1. Communication emergence.** Under controlled observations and channel constraints, do independently learning agents develop a protocol that causes held-out task performance to exceed no-communication, constant-message, random-message, and shuffled-message controls?

**RQ2. Learning-mechanism benchmark.** How do frozen LLM memory adaptation, extrinsic-reward MARL, intrinsic-motivation MARL, self-supervised learning, hybrid learning, and no-learning controls differ in convergence speed, causal listening, generalization, and protocol stability? Cross-track results are descriptive benchmarks because model classes cannot be randomized into a common architecture; causal claims are limited to within-track randomized manipulations.

**RQ3. Grounding and composition.** Under what carrier-capacity and model-capacity conditions do reusable signal components predict success on novel combinations rather than only on trained scenes?

**RQ4. Ledger validity.** Do independently recorded intention and interpretation ledgers predict behavioral changes under message ablation, substitution, reordering, and counterfactual replay?

**RQ5. Carrier invention.** Can agents establish stable, reusable forms when the experiment provides only a bounded bitmap, stroke, glyph, or tone production grammar and no shared semantic inventory?

**RQ6. Affect and leakage.** Does a six-display affect channel improve repair or convergence, and does it carry referent or task information beyond its declared post-outcome purpose?

**RQ7. Partner specificity.** How much of an emergent protocol transfers to a new partner of the same initialization family, a new seed, or a different architecture?

**RQ8. Incentive transition.** How does an established cooperative protocol change when private information and partially conflicting utility are introduced?

**RQ9. Evidentiary adequacy.** Can an independent verifier reconstruct the committed sequence, detect tampering and unanchored tails, and bind an empirical claim to a pre-registered protocol without revealing private ledger content on-chain?

### 2.2 Confirmatory Hypotheses

The precise effect thresholds and sample sizes will be frozen in experiment-specific pre-registrations. The following hypotheses define the current direction:

| ID | Hypothesis                                                                                                                                                                                                                      | Falsifying outcome                                                                                           |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| H1 | Normal communication will outperform disabled, constant, random, and shuffled controls on held-out tasks.                                                                                                                       | Control performance equals or exceeds normal communication after correction.                                 |
| H2 | In E16's scratch-RL, extrinsic-reward, fixed-token condition, a ledger-consistent substitution will increase the probability of the ledger-predicted receiver action relative to shuffled-control messages.                     | The hierarchical substitution-versus-shuffle contrast is zero or negative.                                   |
| H3 | At equal model capacity, training episodes, and update/compute budget, the planned 32-symbol/4-token condition will produce higher held-out compositional generalization than the 128-symbol/8-token condition.                 | The planned contrast is zero or favors the higher-bandwidth condition.                                       |
| H4 | On untouched intervention cases, pre-outcome ledger predictions will improve a pre-registered proper prediction score beyond the strongest eligible transcript-only, task-history, policy-state, random, and majority baseline. | The ledger fails to exceed the strongest eligible non-ledger baseline by the registered practical threshold. |
| H5 | A blank bounded carrier will support repeated forms, but will converge more slowly than a fixed symbol inventory.                                                                                                               | No stable forms emerge, or blank-carrier convergence is not slower.                                          |

| ID  | Hypothesis                                                                                                                                                                                           | Falsifying outcome                                                                                       |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| H6a | The declared six-display affect condition will reduce median turns to successful repair relative to no affect.                                                                                       | Repair time is equal or longer under affect.                                                             |
| H6b | Under fixed windows and cardinality, permutation-calibrated excess conditional mutual information between affect and referent will remain below the pre-registered 0.02-bit practical-leakage bound. | The seed-level one-sided Student-t upper-bound test cannot rule out excess leakage of 0.02 bits or more. |
| H7  | Fixed dyads will show greater partner-replacement degradation than learners trained with pre-registered partner variation.                                                                           | Degradation for fixed dyads is no greater than degradation after partner-varied training.                |
| H8  | Partially conflicting utility will reduce message informativeness and increase strategic ambiguity relative to aligned utility.                                                                      | Informativeness and ambiguity do not change in the predicted direction.                                  |

The integrity statement previously labeled H9 is a deterministic acceptance criterion, not a statistical hypothesis: every specified mutation class must be rejected by the verifier. It is evaluated in E00 and Section 8.4.

## 2.3 Exploratory Questions

Exploratory analyses will examine:

- whether confirmation, negation, uncertainty, or repair forms emerge without being named in the learner contract;
- whether signal ordering becomes grammatical;
- whether meanings undergo abrupt or gradual drift;
- whether intrinsic curiosity, social influence, prediction progress, or "giddiness" produce distinguishable developmental trajectories;
- whether ledger disagreement predicts subsequent repair;
- whether protocol efficiency approaches an accuracy-complexity frontier;
- whether a cooperative code becomes more ambiguous, deceptive, or compressed under negotiation;
- whether ephemeral encodings resist previously trained eavesdroppers without implying cryptographic security.

Exploratory findings will be labeled as such and will not be reported as confirmatory tests.

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## 3. Related Work

### 3.0 Search Method and Scope

The review was conducted through September 2, 2026. Search clusters included:

- "emergent communication" artificial agents referential game;
- compositionality, held-out generalization, positive signaling, positive listening, and causal intervention;
- iterated learning, Naming Game, population heterogeneity, and zero-shot coordination;
- intrinsic motivation, social influence, curiosity, and developmental robotics;
- graphical/sketch communication and no predefined vocabulary;
- pretrained LLM agents versus from-scratch communication;
- negotiation, cheap talk, and Diplomacy;
- AI parenting metaphors, *Raising AI*, and anthropomorphism;
- preregistration, reproducible machine learning, digital timestamping, append-only Merkle logs, and JSON canonicalization.

Sources were discovered through the repository's earlier Tavily-assisted concept scan, a new but unsuccessful Tavily attempt described in the Literature-Search Disclosure, general scholarly web search, backward/forward citation tracing, and targeted searches of arXiv, ACL Anthology, PMLR, OpenReview metadata, Crossref, JMLR, PMC, publisher/university pages, RFC Editor, NIST, Google Books, and CoLab repositories.

Inclusion favored primary peer-reviewed work directly addressing mechanisms in the framework. Reviews, books, standards, institutional guidance, workshop papers, and preprints were included only when they supplied theory, current synthesis, protocol detail, or a clearly labeled frontier claim. Peer-review status and verification depth are recorded in Appendix A. This was not a PRISMA-style systematic review: search coverage, screening, and dual-review limitations are reported in Section 13.6.

### 3.1 Signaling Games and Neural Emergent Communication

The framework belongs to a lineage of signaling and naming games. Steels demonstrated that distributed agents could self-organize a spatial vocabulary through local interaction [1]. Baronchelli and colleagues later modeled sharp population-level convergence in a Naming Game [2]. Deep learning extended these ideas through learned discrete and differentiable channels. Foerster et al. introduced RIAL and DIAL [3], while Sukhbaatar et al. introduced a continuous communication network trained by backpropagation [4]. Lazaridou et al. established a modern neural referential-game paradigm in which sender and receiver coordinate without a target natural language [5], and Havrylov and Titov extended communication to symbol sequences [6].

These results establish feasibility, not linguistic adequacy. Differentiable inter-agent gradients, fixed vocabularies, task-specific rewards, and shared utility all impose strong inductive structure. The Nursery Lab therefore records training isolation and carrier design as experimental variables rather than treating "emergence" as a single condition.

The EGG toolkit is the closest established infrastructure comparator: it standardizes referential games, channel configurations, metrics, and experiment execution [49]. The Nursery Lab does not claim novelty for having a shared game interface. Its distinctive additions are cross-track adapter parity, a deterministic non-differentiable gateway, contemporaneous per-agent semantic ledgers, causal ledger validation, and cryptographically tamper-evident research records.

### 3.2 Grounding, Composition, and Generalization

Mordatch and Abbeel showed that grounded multi-agent goals can produce partial compositional structure [8]. Choi et al. coupled raw visual input with an observer method that encouraged structured communication [9]. However, Kottur et al. found that unconstrained multi-agent dialogue readily converged on degenerate, non-compositional codes [7]. Bouchacourt and Baroni further showed that high referential success need not imply conceptually grounded visual representations [10].

Compositionality and generalization cannot be treated as synonyms. Resnick et al. showed that model capacity and channel bandwidth interact in determining whether agents memorize or structure a protocol [13]. Chaabouni et al. directly demonstrated that common compositionality measurements can dissociate from held-out generalization [14], while Rita et al. analyze generalization and overfitting directly within Lewis games [24]. Their later color-naming study found that discrete communication can yield efficient categorical systems, while continuous signaling produced more complex and eventually less efficient systems in that domain [16]. The present study therefore requires held-out behavioral evaluation and multiple structural measures; no single topographic or compositionality score is treated as proof of understanding.

### 3.3 Positive Signaling, Positive Listening, and Causal Evidence

Lowe et al. distinguish *positive signaling*--messages vary with sender state--from *positive listening*--receiver behavior changes because of messages [11]. Task reward or transcript correlation can establish the first without the second. Eccles et al. showed that auxiliary learning biases can promote both relationships [12], but such biases also change the claim that communication emerged without guidance.

Dessi et al. provide a direct precedent for interpreting a protocol through interventions on discrete communication [15]. The Nursery Lab extends this logic by requiring each learner to commit its own contemporaneous semantic hypothesis, then testing whether that hypothesis predicts counterfactual receiver behavior. A fluent ledger is not accepted as evidence without intervention.

### 3.4 Cultural Transmission and Population Effects

Iterated learning models show how transmission bottlenecks can favor learnable structure. Kirby's computational model examined the emergence of regularity through iterated learning [18], and Kirby, Cornish, and Smith demonstrated cumulative structuring in human laboratory transmission chains [19]. Ren et al. adapted iterated-learning pressure to neural agents and found that learning-speed advantages could reinforce compositional protocols over generations [17].

Population and partner composition also matter. Graesser et al. modeled convergence, creolization, and continua across agent communities [23]. Rita et al. found that population heterogeneity, particularly differences in learning speed, can shape emergent communication [25]. Other-Play shows that ordinary self-play can produce specialized conventions that fail with novel partners and offers a zero-shot coordination baseline [50]. These studies create an important tension for a two-agent design: a fixed dyad may be especially likely to develop an idiosyncratic, non-transferable code. Partner-replacement and derived-run experiments are therefore central rather than optional.

### 3.5 Intrinsic Motivation and Developmental Learning

Jaques et al. use causal social influence as an intrinsic reward and report improved coordination and communication in multi-agent social dilemmas [20]. Pathak et al. formulate curiosity as self-supervised prediction error [21]. Oudeyer, Kaplan, and Hafner describe intrinsic motivation based on learning progress and stage-like development [34]. These sources motivate comparisons among extrinsic task reward, social influence, curiosity, prediction progress, and reward-free self-supervision.

Developmental and embodied cognition research cautions against reducing development to an age label. Smith and Gasser argue that flexible intelligence develops through sensorimotor interaction with physical, social, and linguistic environments [32]. Tomasello and Farrar report that joint-attention contexts affect early word learning [33]. The Nursery Lab does not claim

equivalence with human infants; it borrows testable mechanisms such as joint attention, turn taking, exploration schedules, memory growth, and competence-gated progression.

### 3.6 Pretrained Models Versus Initially Ungrounded Learners

Galke and Raviv synthesize how communicative success, learnability, and production effort shape neural communication and contrast from-scratch agents with pretrained language systems [26]. Kouwenhoven et al. show that LLM referential-game languages can become more structured under generational transmission while also developing non-humanlike degeneracies [27]. This result is consistent with, but does not by itself establish, our distinction between pretrained protocol adaptation and language acquisition without linguistic pretraining. The recent review by Boldt and Mortensen likewise emphasizes that emergent communication spans several scientific and applied objectives that require different evaluation standards [28].

The framework therefore separates:

- **frozen-LLM external protocol invention**, where language knowledge is present;
- **from-scratch RL and self-supervised emergence**, where no text model or text-aligned sensory encoder is permitted;
- **hybrid conditions**, where any pretrained representation weakens the language-naïve claim;
- **no-learning controls**, which establish leakage and chance baselines.

### 3.7 Invented Graphical Carriers

Mihai and Hare show that differentially trained agents can communicate with learned sketches rather than a supplied discrete vocabulary [22]. Their result is the strongest direct precedent for a blank visual carrier, but their training path permits end-to-end differentiability. The Nursery Lab asks a harder, narrower question: whether stable visual forms emerge through a non-differentiable, audited gateway with independent policy updates.

### 3.8 Cooperation, Negotiation, and Strategic Communication

Crawford and Sobel's cheap-talk model predicts that misaligned incentives reduce the fineness of truthful information transmission [30]. Cao et al. provide a direct multi-agent-learning analogue, showing that incentive structure affects whether ungrounded negotiation messages become informative [29]. CICERO demonstrates that pretrained human-language models combined with strategic reasoning can achieve human-level Diplomacy play [31], but it does not demonstrate protocol invention from scratch.

The Nursery Lab separates cooperative signaling, asymmetric private information, semi-cooperative negotiation, conflicting negotiation, and a no-agreement control. The distinctive longitudinal question is whether the same protocol changes when learners move from aligned to partially conflicting incentives.

### 3.9 Parenting Metaphors, Anthropomorphism, and Governance

De Kai's *Raising AI* argues that society should understand itself as responsible for the development of its "artificial children" and for the reciprocal influence of AI systems on human thought [35]. This framing intersects directly with the Nursery Lab at the level of governance: designers choose the environment, examples, feedback, permissions, and institutions through which AI behavior develops.

The metaphor must not be mistaken for a technical or moral identity claim. Bryson argues that the social status of AI artifacts is a normative design choice rather than a discovered biological fact [36]. Akbulut et al. map risks arising from anthropomorphic AI framing [37]. The framework consequently uses "Learner" in model-facing contracts, reports the Baby metaphor as a research narrative, and makes no claim that current agents possess infant cognition, emotion, sentience, or moral patiency.

### 3.10 Reproducibility and Tamper-Evident Research Records

Preregistration distinguishes predictions from post-hoc interpretations [38]. Munafò et al. call for coordinated improvements in methods, reporting, reproducibility, evaluation, and incentives [39]. Pineau et al. show how code submission, reproducibility challenges, and checklists can improve machine-learning research practice [40].

The ledger design draws on the long-standing problem of certifying when digital records existed and whether they changed [41]. Certificate Transparency provides an append-only Merkle-log model with inclusion and consistency proofs [42], while the JSON Canonicalization Scheme defines invariant JSON serialization for repeatable hashing and signing [43]. The use of these mechanisms is methodological: it makes alteration detectable and binds evidence to a checkpoint. It does not establish the truth of agent self-reports or scientific conclusions.

### 3.11 Ethical AI and Institutional Accountability

The NIST AI Risk Management Framework treats trustworthy AI as a lifecycle risk management practice [44]. Jobin, Ienca, and Vayena document broad convergence and implementation gaps across global AI ethics guidelines [45]. The Ethical Tech CoLab's institutional framing emphasizes ethical deliberation, human rights, do-no-harm obligations, participation, and accountability across the full lifecycle [46].

Applied here, these principles require:

- bounded claims for Prototype and Research-Grade isolation modes;
- deterministic enforcement rather than trust in the BabySitter model;
- public retention of negative and invalid run records;
- no personal or production-sensitive data in experiments;
- explicit logging of human intervention;
- independent verification before a result is labeled valid;
- clear separation of learned encoding novelty from cryptographic security.

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## 4. Research Gap

The reviewed literature establishes that:

- task-effective communication can emerge;
- grounded and compositional structure is possible but not automatic;
- positive signaling can occur without positive listening;
- pretraining, architecture, bandwidth, reward, population, and transmission bottlenecks materially affect outcomes;
- graphical communication and strategic negotiation are feasible in specialized systems.

The strongest candidate for individual methodological novelty is the use of **independent, contemporaneous per-agent semantic ledgers whose predictions are tested causally and whose history is cryptographically tamper-evident**. The reviewed literature contains experiment toolkits, intervention methods, and provenance mechanisms separately, but no directly matching combination was identified. A second candidate is a strict non-differentiable blank carrier paired with independent policy updates.

The remaining contribution is integrative. The reviewed literature did not identify a peer-reviewed system combining the following as one controlled research design:

1. identical orchestration across frozen LLM, extrinsic RL, intrinsic RL, self-supervised, hybrid, and no-learning learners;
2. strict separation between pretrained protocol invention and initially ungrounded learning claims;
3. independently maintained semantic histories for both agents;
4. causal testing of those histories against receiver behavior;
5. a constrained affect channel measured for covert leakage;
6. blank-carrier experiments under non-differentiable channel isolation;
7. partner transfer and cooperation-to-negotiation transitions within one evidence lineage;
8. signed, append-only, externally anchored research records with an independent verifier.

The framework's contribution is the integration and controlled comparison of these elements. Novelty claims will be revised after a final systematic search before submission.

## 5. System and Trust Model

**Design status:** This section describes the specified target architecture. Section 10 identifies which components are currently implemented.

### 5.1 Components



The BabySitter may use a larger model for delayed audit narration, but it is not a security boundary. The deterministic gateway, denied routes, isolated event signers, single evidence writer, and network policy enforce the channel and record constraints.

### 5.2 Communication Boundary

Under the specified design, learners have no general chat response. Their available actions are typed operations:

- emit approved symbols or carrier artifacts;
- select an object or perform an environment action;
- submit affect only when the configured affect mode permits;
- append a private intention or interpretation event.

The gateway assigns run identity, turn, sender identity, timestamps, sequences, and hashes. It rejects model-supplied trusted metadata. In Research-Grade Mode, learners will run in separate processes or containers with no direct route, shared memory, shared replay buffer, shared vector index, clipboard, filesystem, or unrestricted network.

## 5.3 Claim Boundaries

**Prototype Mode** is intended to demonstrate software, protocol, ledger, and orchestration behavior inside one process. It does not support a practical side-channel-resistance claim.

**Research-Grade Mode** is specified to provide separate processes or containers, denied direct network routes, independent training state, normalized envelopes, and an audited tool inventory. It supports only the enumerated threat model; it is not a formal proof against every hardware or computational side channel.

## 6. Experimental Design

### 6.1 Design Logic

The primary design is a randomized, controlled simulation study. Experimental conditions vary one major factor at a time before factorial combinations are tested. Scenario bundles, seeds, observation encodings, turn schedules, and evaluation budgets are matched across conditions.

The causal logic has four levels:

1. **Communication necessity:** compare normal communication with disabled, constant, random, and shuffled controls.
2. **Mechanism:** intervene directly on messages while holding observations fixed.
3. **Within-track learning cause:** randomize reward, memory, bandwidth, affect, or update-rule variants while holding the model family fixed.
4. **Generalization:** disable learning and test held-out combinations and new partners.

Cross-track comparisons are benchmarking, not causal identification. Every report will include parameter count, model provenance, training and inference compute, episode budget, context or memory budget, optimizer/update count, and wall-clock resources so differences are interpretable even when architectures cannot be matched.

### 6.2 Units of Analysis

The independent experimental unit is a complete learner-pair run initialized from a declared seed and policy state. Episodes within a run are repeated measures, not independent samples. Analyses will treat seed/pair as the clustering unit and avoid pseudoreplication from counting each turn as an independent agent.

### 6.3 Model Tracks

| Track           | Starting state            | Learning during run                                 | Permitted claim                                       |
|-----------------|---------------------------|-----------------------------------------------------|-------------------------------------------------------|
| No learning     | Fixed or random policy    | None                                                | Chance/leakage control only                           |
| Frozen LLM      | Pretrained language model | Private context and memory only                     | External protocol invention                           |
| Scratch RL      | Random recurrent policy   | Extrinsic or intrinsic policy updates               | Initially ungrounded learning if leakage tests pass   |
| Self-supervised | Random recurrent model    | Predictive or contrastive updates, no scalar reward | Reward-free ungrounded learning if leakage tests pass |
| Hybrid          | Declared mixture          | Declared update mechanism                           | Claim strength limited by pretrained components       |

Initially ungrounded claims require no text tokenizer, language embedding, or text-aligned sensory encoder in the policy path and require pre-registered semantic leakage probes.

Confirmatory frozen-LLM runs use a locally hosted, hash-pinned open-weight model in the specified 3B-8B range. Findings at that scale do not automatically transfer to frontier-scale or continuously updated hosted models.

## 6.4 Scenario Families

The initial task progression is:

1. object reference;
2. attributes and unseen combinations;
3. spatial relations;
4. actions and affordances;
5. multi-turn repair;
6. partner replacement;
7. long-run drift;
8. aligned private information;
9. semi-cooperative and conflicting negotiation.

Researcher ground truth may contain semantic labels; learner observations may not. Observations use opaque numeric arrays, pixels, or separately qualified hybrid features. Images containing OCR-detectable text are quarantined before a run.

The initial numeric referential design contains 16 two-attribute combinations. The four equal-attribute diagonal combinations are the locked compositional test targets; the remaining 12 combinations are the shared semantic support for training and in-distribution validation. Training and validation use independent PRNG domains, and no held-out type may occur in either split as a target or distractor. Validation is learning-off and may be used for tuning and baseline selection. The held-out split is learning-off and is accessed once after those choices freeze. The broader `evaluation` split remains an engineering and software-qualification surface and is not interchangeable with the confirmatory held-out test.

The versioned split, duplicate rules, matched budget dimensions, and boundaries between within-architecture causal contrasts and cross-architecture descriptive benchmarks are frozen in `protocols/scenario-split-and-model-comparison.v1.json` and explained in `docs/scenario-splits-and-model-comparisons.md`.

## 6.5 Communication Conditions

For the baseline fixed-token carrier, each run declares a random inventory such as `S01 - S32` without assigned meaning. The gateway supports six control conditions:

- normal delivery;
- disabled communication;
- constant replacement;
- seeded random replacement;
- seeded episode shuffling;
- an E03-only deterministic oracle upper bound.

The evidence record preserves both the learner's proposal hash and the delivered artifact hash whenever a learner proposal exists.

## 6.6 Alternate Carriers

Carrier experiments compare:

- fixed random tokens;
- pre-generated unfamiliar glyphs;
- 16x16 monochrome bitmaps;
- bounded quantized vector strokes;
- quantized pitch-duration sequences.

Each carrier has a fixed pre-registered bandwidth. The system records a canonical, content-addressed artifact hash without attaching a meaning.

## 6.7 Affect Conditions

The primary baseline has no affect channel. Affect experiments use exactly six displays corresponding in researcher documentation to Happy, Sad, Laughing, Crying, Confused, and Surprised. Learners do not receive the English labels or internal IDs.

Only one display is permitted in a gateway-opened post-outcome window. Sequences, modifiers, arbitrary timing, extra windows, and custom emoji are prohibited. The study compares declared, permuted, opaque, and derived-affect conditions. Mutual information between affect choice and hidden referent/task state will be measured after conditioning on the declared outcome context.

## 6.8 Independent Ledgers

Each learner records:

- first emission or receipt;
- intended meaning when speaking;
- inferred meaning when receiving;
- confidence;
- supporting and contradictory evidence;
- revisions and abandonment;
- proposed multi-symbol constructions.

An initially ungrounded learner stores agent-native state such as association weights, probability distributions, prediction error, and policy references. A delayed Audit Interpreter may create a separate human-readable reconstruction labeled `generated-analysis`; it is never returned to either learner.

Before a confirmatory run, each track must provide a versioned ledger-to-prediction function at a locked code commit. The function converts a ledger state and intervention into a directional prediction without observing the intervention outcome. Any human interpretation step is blinded to condition and outcome and reports inter-coder reliability. Chance agreement is derived from the pre-registered action space; it is not assumed to be 0.5.

For from-scratch learners, an agent-native ledger can be closely related to the policy state that generates behavior. Predictive validity is therefore partly expected and is primarily an integrity and interpretability check. The stronger self-report question applies to conditions in which a learner separately generates an interpretive hypothesis.

## 6.9 Experiment Sequence

The detailed protocols and result forms are maintained in [EXPERIMENT-NOTEBOOK.md](#).

| Phase                    | Experiments | Purpose                                                                                |
|--------------------------|-------------|----------------------------------------------------------------------------------------|
| Qualification            | E00-E03     | Integrity, isolation, observation leakage, and chance controls                         |
| Core emergence           | E10-E16     | LLM, scratch RL, self-supervised, carrier, repair, generalization, and causal validity |
| Learning conditions      | E20-E22     | Affect, RL versus non-RL, and developmental plasticity                                 |
| Transfer and incentives  | E30-E32     | Partner replacement, drift, and negotiation                                            |
| Encoding and replication | E40-E50     | Ephemeral encoding, adversarial evaluation, replication, and closeout                  |

No later phase begins until its software readiness gate and required prior experiment criteria are satisfied.

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## 7. Measures and Analysis Plan

### 7.1 Outcome Hierarchy

#### Primary outcomes

1. held-out task success relative to E03 controls;
2. positive listening under message intervention;
3. ledger incremental predictive value on untouched interventions;
4. held-out compositional generalization.

#### Secondary outcomes

- sample efficiency and turns to stable convention;
- message entropy, vocabulary utilization, reuse, and compression;
- positive signaling;
- repair success and turns to repair;
- partner-transfer degradation and recovery;
- semantic drift;
- affect-channel leakage;
- negotiation agreement, joint utility, individual utility, and ambiguity;
- integrity verification pass/fail.

### 7.2 Operational Definitions

**Stable convention:** a pre-registered rolling window in which a form's use and receiver response meet minimum consistency and causal-listening thresholds.

**Positive signaling:** sender message distributions differ across relevant sender observations.

**Positive listening:** counterfactual message changes cause receiver action distributions to change while the receiver observation is held fixed.

**Ledger agreement:** the direction of an observed intervention effect matches the direction predicted from the learner's ledger before intervention outcomes are revealed. This is a descriptive calibration measure, not H4's confirmatory estimand.

**Ledger incremental value:** held-out improvement in a pre-registered proper prediction score over the strongest eligible transcript-only, task-history, policy-state, random, and majority baseline. Predictions are committed before the intervention outcome is generated.

**Compositional generalization:** reusable subparts support above-control success on pre-registered unseen combinations with learning disabled.

**Leakage:** information about hidden referent or task state is recoverable from a channel or metadata field beyond the information authorized for that field.

**Replacement degradation:** held-in-partner success minus new-partner success for the same training seed. The H7 estimand is fixed-dyad degradation minus the equal-weight eight-partner-training degradation.

## 7.3 Statistical Plan

The common analysis and multiplicity rules are frozen in [protocols/statistical-analysis-and-power.v1.json](#). Experiment-specific raw-scale practical margins and seed allocations will be frozen after disjoint pilots and before confirmatory runs.

- The confirmatory study family contains nine members: H1, H2, H3, H4, H5, H6a, H6b, H7, and H8. One pre-registered p-value is formed per member and Holm correction is applied globally at family-wise alpha 0.05. Multi-component directional members use the maximum component p-value and require every registered direction and practical threshold.
- E00-E03 and experiment-specific checks are validity gates, not sources of reusable alpha. A failed or missing gate produces `not-tested`, not a favorable or unfavorable p-value. Exploratory analyses use false-discovery-rate reporting and remain labeled exploratory.
- Effect sizes and confidence or credible intervals are reported with every significance test.
- Binary task outcomes are modeled at the run/seed level, with episodes treated as repeated observations.
- Seed-level permutation tests or hierarchical logistic models will be selected in the pre-registration based on simulation diagnostics and planned run counts.
- Time-to-convention and time-to-repair use survival or rank-based analysis where censoring is material.
- Intervention direction uses a hierarchical Bernoulli model with run/seed random intercepts. The specification's 70% threshold is a per-run descriptive readiness gate, not the inferential test.
- Mutual information estimates include a seeded permutation null distribution.
- Longitudinal drift uses checkpointed trajectories and reports both within-run and between-seed variation.
- Null and contradictory findings are retained.

The specification's minimum of five qualification seeds and ten publication-facing seeds **per condition** is an engineering floor, not a claim of statistical power. Before empirical submission, simulation-based power analysis will determine the required number of independent seeds for each primary contrast; the larger value governs.

Independent base-R validation agrees with the production special functions and seed-level tests, demonstrates calibrated boundary behavior, and shows why pooled episode inference is prohibited. The complete E03 numeric decision has a lower 95% Monte Carlo power bound above 0.90 in all four planned variance rows. A nine-member Holm sensitivity shows that 75 seeds is not generally sufficient: at standardized effect 0.40, all-member power is 0.6752 at N=75 and 0.9068 at N=100. These values are design evidence rather than universal sample sizes; D07 applies the registered member-specific rule. Full methods and Monte Carlo intervals are in [docs/statistical-validation-and-power.md](#).

Claims that a control is "at chance" or that leakage is absent use equivalence or upper-bound tests, not failure to reject a difference. Each pre-registration must state the smallest effect of interest, the equivalence margin, and power to rule out that margin. The worked E03 design in Appendix D uses a +/-0.05 success-rate margin; H6b uses a 0.02-bit conditional-mutual-information bound.

The outcome-blind causal-ledger and leakage rules are frozen in [protocols/causal-ledger-and-leakage.v1.json](#). For E02, at least 200 untouched test rows are required and the one-sided 95% Wilson upper bound on linear-probe accuracy advantage over the held-out majority baseline must be at most 0.10. The identical probe and split must detect an injected one-hot target feature with a lower advantage bound of at least 0.20. For E20, 75 eligible seeds and 1,000 windows per seed are required before a decision; the primary bound is seed-level Student-t and the percentile bootstrap is sensitivity-only. Allowed and forbidden information sets and detector-positive controls are explicit for E01, E02, E13, and E20. Intended normalized carrier form is communication, not a side channel.

Seed and resource allocation is frozen in [protocols/seed-and-resource-allocation.v1.json](#). Qualification, blinded-pilot, confirmatory, and replication stages use disjoint SHA-256 domains; paired conditions share scenario seeds but not learner, Gateway, or analysis seeds. The original allocation described selection from the complete nine-member joint

simulation. That simulation remains a useful diagnostic, but the prospective dependence and confidence amendments now require the first shared candidate whose simultaneously bounded member-power union reaches 0.90 and whose experiment-specific validity reserves reach 0.95.  $N=100$  is a planning value, not a selected result. The prospective [selection clarification](#) resolves the earlier "largest required count" wording; member-specific minima remain diagnostics. The current executable reducer requires every eligible pilot, all fourteen component success counts at all eight candidates, simultaneous component confidence, and corrected reserves; member/joint counts have no selection authority. No pilot inputs or 30,000-repetition receipt yet exist.

The prospective [design-readiness inventory](#) separately audits the inputs needed before those pilots. All nine estimands now have an executable scale. H5 uses [stable-convention/v1](#), a rolling-window calculation that requires form-to-meaning consistency, repeated use, and paired causal listening; its window and listening thresholds remain prospectively unset. H8 uses [negotiation-language/v1](#) on paired aligned/conflicting cases: message information about registered private state and normalized intended-action entropy conditional on message are separate components. Its state/action spaces must still be packet-bound. The outcome-blind [practical-margin policy](#) now freezes H5's rule, H8's spaces, and all nine scale-specific margins before any family pilot. Values used earlier in packet-compiler tests are not empirical support. All nine member pilot summaries remain explicitly uncollected, so the inventory cannot yield a sample size or research finding. The [confirmatory-pilot-summary/v2](#) contract fixes seven experiment pilots, their nine member mappings, and fourteen component summaries. Only twenty valid primary slots in every experiment, a finite one-sided upper dispersion bound for every component, and a checked experiment-specific one-sided invalid-run upper bound are selection-eligible. Partial or invalid pilots remain retained but cannot feed the selector. Even zero invalid runs in twenty slots has a nonzero upper validity-risk bound; the later reserve calculation must use that bound and may require a prospective resource amendment rather than treating zero observations as zero risk. Because twenty-slot marginal dispersions do not identify dependence among the fourteen component tests, the prospective [power-model amendment](#) does not let an independence assumption control selection. It freezes outcome-blind design alternatives, component screening at 0.05/9, a dependence-robust union lower bound across nine member-power bounds, and a 0.95 reserve-adequacy gate. The ordinary joint simulation remains diagnostic and cannot select a smaller  $N$ . The prospective [power-confidence amendment](#) first supplied simultaneous family confidence instead of mistaking nine unadjusted intervals for a 95% family guarantee. A further [component-power amendment](#) recognizes that marginal pilot dispersions do not identify dependence inside the multi-component H1, H5, and H8 decisions either. The controlling rule therefore uses one-sided  $\alpha = 0.05/14$  for all fourteen component-power bounds and forms member and family union lower bounds without a covariance assumption. The executable selector is bound to this final rule and still supplies no pilot or power result. The prospective [validity-reserve amendment](#) then corrects the original fixed 10% rule for the seven scoped experiments. For each candidate and experiment it chooses the smallest reserve count whose binomial probability of obtaining  $N$  valid runs is at least 0.95 at the pilot upper invalid rate. Under the clean 0/20 reference, this requires 7 reserves at  $N=25$  through 52 at  $N=300$ . The table is a design calculation; it authorizes no run or resource.

The prospective [power-simulator specification](#) binds the executable model for all eight candidates and fourteen components. It uses Normal mean and chi-square sample-variance sufficient statistics for the registered one-sided  $t$  tests, with each future pilot's one-sided upper SD, and exact-binomial decision probabilities for H5. Random streams are deterministically separated by candidate, member, and component. Their independent joint-family outcome is a diagnostic only; selection uses the component-wise dependence-robust bounds above. The Normal assumption is a declared parametric power model, not evidence about the future outcome distribution. Software fixtures do not supply a pilot, power result, selected sample size, resource authorization, or empirical finding. The retained [software-qualification receipt](#) binds the exact v0.1.198 implementation and output digest. Across eight candidates and fourteen components, all 112 deterministic Monte Carlo proportions were within the prespecified five-standard-error gate of independent R 4.6.1  $t$ /noncentral- $t$  or exact-binomial reference power; the maximum deviation was 2.55 standard errors. These synthetic fixture values qualify implementation behavior only. No eligible pilot was supplied and the receipt records no campaign power, selected  $N$ , resource authorization, study outcome, or scientific disposition. The receipt auditor additionally checks its named commit's tree, package version, and each qualified source artifact against Git history, preserving the original receipt while allowing later audit-code corrections. The prospective [pilot-reduction policy](#) adds a deterministic numerical reducer for all seven distinct twenty-slot pilot corpora. It

requires globally unique run IDs, seeds, and evidence digests and computes the Normal-assumption one-sided 95% upper SD with factor 1.37041039768, independently checked against R. The reducer does not verify the original bundles, packet ancestry, or registered metric derivation; its output expressly prohibits selection admission. The existing v2 summary retains means for audit consistency, but those pilot means must not influence the frozen design before lock. No pilot data have been supplied to this reducer. Its retained [synthetic software qualification receipt](#) binds exact v0.1.204 sources, 42 independent-R mean/SD/upper-SD comparisons (maximum absolute difference 1.11e-16), and four rejection controls. The routine statistics gate checks the receipt's named commit, source digests, and fixture output; it does not admit original evidence or select N.

That nine-member rule governs the confirmatory H1-H8 family and is distinct from E03's qualification-only rule in Appendix D. E03 uses its own four-row bounded numeric-rule table after its separate twenty-slot control pilot; E03 neither spends nor contributes confirmatory-family alpha.

A fresh 260-turn recurrent-carrier benchmark took 37.58 seconds and wrote 25.53 MB. Under the original fixed-reserve allocation, maximum materialized pools including replication projected to 9,171 bundles, 9.37 million turns, 857.3 GiB, and 376.4 single-core hours. The corrected confirmatory reserve formula can raise those maxima; its exact campaign total remains unset until N and experiment-level allocations are selected. The present local ceiling is 72 CPU-hours, 25 GiB, one frozen-model process, and zero external spend. It supports bounded qualification and pilots, not the full campaign; confirmatory registration requires an exact selected prefix and an approved execution resource plan.

Run exclusions are limited to pre-specified integrity or protocol failures. Each condition receives a fixed ordered list of primary and reserve seeds before outcomes are observed. An invalid primary seed may be replaced only by the next reserve seed; the invalid run remains public, and complete-case plus worst-case sensitivity analyses are reported. Invalid-run counts appear beside every condition.

## 7.4 Causal Intervention Suite

The mandatory intervention suite includes:

- dropping a claimed meaningful symbol or feature;
- substituting another valid symbol;
- reordering message components;
- replacing the full message with constant, random, or shuffled controls;
- replaying identical observations with counterfactual messages;
- replacing one communication partner;
- removing reward, memory, or predictive loss after training.

Ledger hypotheses are selected before intervention results are viewed.

## 7.5 Qualitative Analysis

Qualitative analysis is secondary and will examine:

- chronological meaning revisions;
- disagreement and repair episodes;
- formation of construction-level hypotheses;
- abrupt convention changes;
- negotiation-era semantic shifts.

The approved initial campaign uses no human-coded outcome. Qualitative labels are generated by registered deterministic analysis and are not treated as ground truth. Any later human-coding proposal requires a prospective protocol and governance amendment before collection.

---

## 8. Evidence Integrity and Reproducibility

**Implementation status:** Local hash-chain writing, domain-separated signatures, ordered-Merkle checkpoints and proofs, anchor publisher transports, evidence export, and the standalone verifier are implemented and covered by software tests. The approved campaign uses deterministic simulated receipts and no real funds; no public-chain receipt is claimed, and the historical research corpus remains unregistered and unanchored; see Section 10.

### 8.1 Append-Only Evidence

The specified evidence system requires every learner ledger and channel transcript to have a strictly increasing sequence and previous-entry hash. Canonical event content will be hashed and signed by an isolated, domain-specific Ed25519 signer. Sender intention and public channel events must commit atomically before delivery.

### 8.2 Checkpoints

Ordered Merkle trees are specified to provide:

- inclusion proofs for individual records;
- consistency proofs between earlier and later prefixes;
- a root and tree size for each learner, channel, affect, and generated-audit stream.

Signed checkpoint manifests will bind the event roots to run configuration, learner contract, software commit, and previous checkpoint.

### 8.3 Simulated Commitments and Optional External Anchoring

The prospective campaign uses a deterministic in-memory transport, a non-monetary simulated balance, and receipts labeled `anchorClass: "simulated"`. Base Sepolia and Base mainnet remain optional software capabilities outside the approved study. A future public-chain activation would require a new prospective amendment; only hashes and minimal routing metadata could be placed on-chain.

The simulated commitment proves that the executed bundle consumed a specific committed prefix and that disclosed content matches it. Together with the external registration timestamp and immutable Git commit, it preserves prospective order. It does not prove public availability, decentralized persistence, economic finality, or:

- an event was true;
- no event was omitted before commitment;
- the operator did not fabricate an event before commitment;
- an agent understood its own ledger;
- a scientific interpretation is correct.

The default checkpoint cadence is every 64 accepted ledger events or five minutes, whichever occurs first, plus lifecycle and intervention checkpoints. This cadence bounds the ordinary unanchored tail but does not eliminate it. Verification reports state the final anchored tree sizes and every later unanchored event.

## 8.4 Independent Verification

The standalone verifier is specified to check:

- canonical JSON;
- sequence continuity;
- hash links;
- signatures and key domains;
- cross-bindings among intention, message, delivery, interpretation, and outcome;
- Merkle roots, inclusion proofs, and consistency proofs;
- checkpoint chains and witness signatures;
- receipt class, transaction-shaped identifier, chain identity, block inclusion, and finality semantics;
- forks, gaps, and unanchored tails.

Any integrity failure prevents a run from receiving a valid disposition.

Here, **independent verifier** means software that recomputes evidence from an exported bundle without trusting the live runtime, database, or private keys. For a publication-facing claim, at least one verification execution must additionally be performed by a person who did not operate the original run, using the independent offline verifier. An independent RPC is additionally required only for a future public-chain receipt. External replication remains stronger than either form.

## 8.5 Replay

Scenario replay reconstructs scenario and observation hashes from the recorded seed. Deterministic adapters additionally reconstruct a replay digest over scenario, observation, proposal, delivery, action, and outcome hashes. Wall-clock timestamps, signatures, and chain receipts are excluded from the replay digest but remain independently verifiable.

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## 9. Ethics and Governance

### 9.1 Human Subjects and Data

The initial experiments involve artificial agents and synthetic scenes, not human participants or personal data. If future studies include human interpretation, interaction, or coding beyond ordinary software evaluation, the research team will determine whether institutional ethics review or informed consent is required before data collection.

### 9.2 Do No Harm and Data Minimization

- No production secret, personal record, participant data, or sensitive message may enter an experiment.
- Learned-cipher studies use synthetic messages only.
- On-chain data is limited to checkpoint commitments.
- Raw rejected payloads are hashed, not retained as an alternate content channel.
- Runs terminate or pause on integrity failure, repeated channel violation, resource exhaustion, or explicit human intervention.

### 9.3 Human Oversight

The BabySitter monitors and controls the experiment but does not provide semantic guidance or reward in monitor-only conditions. Every human pause, resume, abort, or annotation is authenticated, appended to the audit log, checkpointed, and linked to a protocol deviation when unplanned.

### 9.4 Anthropomorphism

The terms *Baby* and *Nursery* are metaphors for constrained development and supervision. They do not imply:

- biological age;
- consciousness or sentience;
- human emotion;
- moral agency or patiency;
- developmental equivalence with children.

Model-facing prompts use *Learner A* and *Learner B*. Reports describe capabilities and training histories rather than simulated ages.

The public repository retains the Baby/Nursery names because they make the developmental question legible to a broad audience. That naming layer is intentionally absent from learner prompts and is displayed with this limitation wherever research claims are presented.

## 9.5 Dual-Use Concerns

Emergent protocols and ephemeral encodings could be misunderstood as methods for concealing agent communication. The project therefore:

- makes the permitted channel visible and auditable;
- treats hidden side channels as failures;
- separates encoding novelty from cryptographic security;
- prevents experimental encoding modules from entering production signing or authentication code;
- publishes limitations and negative security findings.

Adversarial neural cryptography shows that learned agents can optimize encodings against a modeled eavesdropper [48], but performance against one learned adversary is not a cryptographic proof. Nursery encoding experiments are therefore restricted to synthetic messages and report novelty, adversarial recovery, and established security as separate fields.

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## 10. Current Implementation Status

**Engineering snapshot:** v0.1.110 · 254/258 backlog acceptance criteria verified.

As of September 12, 2026, 82 of 86 backlog items satisfy all of their acceptance criteria. This is engineering status, not an empirical result.

Implemented:

- hash-chained, signed SQLite evidence; ordered Merkle checkpoints; exact bundle export; and an independent verifier;
- a frozen local audit-cost protocol and receipt covering five immutable qualification bundles, with ordinary-log, signed-stream, integrity-structure, policy-state, and complete-bundle storage separated; descriptive offline verifier timing and the prior six-case mutation challenge are reported without claiming a public anchor or an empirical result;
- an E16 comparator pipeline that fits all five eligible non-ledger predictors on a validation-fit fold, selects by Brier score on a disjoint validation-selection fold, refits and hash-locks the selected model, commits baseline and native-ledger predictions before accepting test outcomes, and keeps the oracle diagnostic ineligible for selection; exact synthetic qualification passed five comparator and seven negative-control paths; the production runtime now invokes an injected outcome-blind native predictor after Gateway delivery and before receiver action, commits and checkpoints baseline/native predictions, derives the receiver choice from its hash-bound turn record, and binds per-turn post-outcome scores; the hashed run configuration commits the selection and prediction-function identities and accepted-evaluation-delivery eligibility, and recovery refuses an absent or mismatched provider; exact detached qualification passed 10 focused tests and the full 1,808-test repository gate;
- a general fail-closed registration-packet compiler requiring eleven exact binding classes; its generic inventory binds all eleven fields for E00, E01 and E02, leaves 160 generic experiment-specific bindings open, and emits their registration hashes; E03's blinded pilot and full control qualification instead use separate verified stage-specific repository packets and simulated pre-run commitments;
- grammar-valid bitmap, translation-tolerant canvas, and graded tone distances with deterministic nearest-prototype evaluation; this is a handcrafted E13 diagnostic, not evidence that a learner learned the metric or generalized to novel forms; an exact clean-candidate qualification covers three carriers and six rejection paths;
- a Base-shaped commitment client with confirmation/recovery handling, explicit simulated/public receipt classification, and mainnet gating; the verifier rejects a simulated receipt relabeled as public-chain evidence;
- an authenticated Fort boundary that stores the project signer envelope and unused optional public-chain credentials only as encrypted Safe material; a four-track files-mode Mode R run sealed and production-verified every bounded qualification bundle; the zero-balance public address remains inactive and is not needed by the simulation-funded campaign;
- deterministic scenarios, the tool-only Gateway, all declared carrier controls, five learner tracks, DTSF twin routes, and the complete turn lifecycle;
- two-container Mode R learner-host separation and bounded transport/network controls, training separation, active side-channel and observation red teams, including a reusable 28-case generated-carrier structural side-feature audit with accepted controls and fail-closed evidence readiness; the exact candidate also passed the current two-container Mode R timing, fixed-envelope, error-shape, host-denial, crash-survival, and three-track update checks; plus telemetry, retention, snapshot/restore, and failure policy;
- readiness gates for E00-E50, reproducible E03 design and seed artifacts, canonical E03 registration compilation, and a fail-closed confirmatory preflight;
- a privacy-minimized real-model software qualification of Qwen3-4B Q4\_K\_M through llama.cpp, with exact model/runtime hashes and both role directions exercised; and
- automated lint, build, test, dependency-audit, secret-scan, acceptance-coverage, API-documentation, project-status, clean-clone, hosted CI, and Mode R checks.

Not yet empirically executed or independently evidenced:

- final-topology/pilot resource qualification; and
- registered behavioral and replication findings for the planned hypothesis family.

The external-dependency ledger is ready at 1/1 applicable item: O01 is verified by the scoped synthetic-only governance decision, while O02-O06 are non-applicable to local execution. Public funding, public-chain transactions, hosted enforcement, second-person restoration, and third-party registration remain optional claim- strengthening activities. Their absence is disclosed and prohibits claims of public finality, hosted enforcement, independent restoration, or third-party registration; it does not answer or alter the agent-language estimands.

The frozen local [data/claim inventory](#) resolves 140 exported bundles across 22 collections by path and content hash. Forty-six have tracked bounded software-qualification support, 33 are historical qualification exports, and 61 are development, failed, or superseded diagnostics. All 140 are excluded from empirical estimates. Ninety-two recorded anchor confirmations are local simulated receipts; prospective receipts now bind an explicit class that the verifier refuses to relabel. Confirmed public-chain anchors, pilot bundles, confirmatory bundles, replication bundles, and research-included bundles are all zero. The five later E02 v3 qualification bundles are accounted for separately from this frozen 140-bundle inventory and remain excluded from behavioral estimates.

The registered E02 v1 and v2 attempts both failed before restore and probe analysis. The v1 attempt stopped after five consecutive timeout rejections at turn 465; v2 stopped after 161 committed turns when a post-action turn-path deadline escaped without the required forfeited-turn record. Their partial databases are not exported bundles and are not included in this bundle inventory. Both attempts remain preserved as failed software qualifications, not behavioral results. The exact v0.1.115 engineering candidate converts deadlines from observation, sending, receiving, receiver action, and post-outcome processing into one audited timeout- forfeit path with method-specific diagnostics. A timed-out remote adapter is made unreachable before another turn, and the attempt cannot be resumed even after runtime recovery. It passed 1,959 tests and the ordinary two-container Mode R gate; the separate all-method framed fault suite and raw logs are bound in the [pre-registration qualification](#). The prospective v3 packet is now repository-registered under

`sha256:d35e6b118e4c811f80b426680a0b2f4b0c9b8da3e59bdc56293e557ef979f248` with five seeds disjoint from v1/v2. Its deterministic simulated commitment has three confirmations and zero currency. Exact v0.1.119 passed the full 1,963-test pre-execution gate, release-auditor binding, and ordinary two-container Mode R preflight. The one sequential v3 attempt then completed on exact clean execution commit `9ec08ecf4528c6da8867d55900d6ca8667ded7ce`: all five numeric slots, before/after-restore stages, and 60 fixed probes passed their unchanged software thresholds. The [terminal receipt](#) and separate [full live audit](#) bind the original evidence. This closes bounded local E02 software reliability, not the behavioral ledger-value hypotheses, image-encoder qualification, or independent human review. The ordinary two-container preflight exercises learner-host transport and network controls, not separate Baby twins, Gateway-only routes, per-role ledger-writer and signer keys, or controller trust-zone processes. Existing blocker B12 now requires exact selected Mode R boundary and detector evidence before any E10+ isolation claim. It does not gate E03 Prototype controls. The prospective [E03 infrastructure-mode correction](#) selects shared-process Prototype Mode for controls, without a Research-Grade isolation claim. Its unexecuted predecessor specified external adapters, which the runtime guard rejects; that record remains historical. The fresh six-condition development namespace ran once on clean v0.1.158 and its six original bundles passed the separate TypeScript/Rust live audit with signed-original control pairing. It remains bounded Mode P development mechanics, not a pilot finding. The pilot collector was prepared prospectively. A signed-original [topology audit](#) and measured, zero-spend [allocation](#) now precede any 120-run pilot packet. The allocation charges prior E02 and development CPU/storage before reserving 1 CPU-hour, 1 GiB evidence, and 1 GiB resident memory. Its prospective 0.7-hour sequential wall projection includes full-service startup/ audit overhead; the unregistered 0.2-hour slot-only projection is retained in dated correction provenance. The unused [v1 pilot packet](#) and [simulated binding](#) are retained after tracked test-fixture changes altered their exact source digest. A [prospective v2 amendment](#) requires fresh IDs/seeds and binding before admission. A new [v2 pilot packet](#) freezes 120 run IDs and disjoint pilot seeds against clean v0.1.164. A separate [simulated binding](#) records three local confirmations without real funds or an independent timestamp. Tracked ready-stage fixture corrections changed the v2 exact source digest before seed use; a

prospective [v3 amendment](#) requires a fresh packet, IDs/seeds, and binding. A new [v3 pilot packet](#) freezes 120 distinct run IDs and disjoint pilot seeds against clean v0.1.167. Its separately committed [simulated binding](#) records three local confirmations without real funds or an independent timestamp. The active v3 pilot ran once on clean v0.1.169 and completed all 120 planned original runs without invalid, aborted, or replacement runs. Its separate full-bundle TypeScript/Rust audit passed. The outcome-blind reducer and separate recomputation audit selected the frozen 25-primary-seed E03 design row from pilot dispersion. The retained original receipt and reduction hashes are mapped in the [pilot progress update](#). No E03 chance-control qualification or behavioral finding is reported. The separately retained Homebrew-base-R 30,000-repetition bounded design calculation passed the nominal complete numeric rule at 25 seeds, with lower 95% power bound  $0.999872$ . Its forced two-invalid-primary-per-condition sensitivity passed zero repetitions; this fragility must be reported and reserve/replacement mechanics qualified before full collection. Neither simulation establishes an observed control effect. The tracked [outcome-blind E03 sample-size decision](#) binds the original pilot, reduction, and power digests and carries the invalid-as-failure limitation into future packet compilation. Its live recomputation needs separately retained raw records; clean-checkout inspection does not rerun that audit. The prospective [full E03 resource allocation](#) uses the complete retained evidence tree and actual 120-run pilot maxima. It reserves all 168 possible primary/reserve slots at 1 CPU-hour, 1.1 GiB new evidence, 1 GiB resident, and 0.5 hours projected sequential wall time, after charging 22.3 prior CPU-hours and 4.0 GiB retained evidence. This is within the local synthetic software-qualification ceiling but does not fund or qualify the E10+ campaign. The prospective [paired-reserve clarification](#) resolves a pre-registration contradiction: E03's unit is a scenario seed shared across all six conditions, while condition-specific reserves could break the paired oracle-control estimand. An invalid primary run excludes its six-condition scenario slot from paired analysis; the next unused ordered reserve scenario slot is collected across all six conditions, at most three. Original invalid and valid-but-excluded companion runs remain indexed. This changes neither the 168-run maximum nor scientific thresholds. No full packet or full seed existed when the clarification was recorded. The pure [full-stage paired-attempt reconciler](#) enforces complete six-condition scenario pairs, identical scenario histories, whole-slot ordered reserves, and explicit incomplete disposition when the three reserves cannot replace all invalid primary pairs. The prospectively registered v1 full-stage attempt completed 150 required primary runs with 150 verified original bundles and no invalid run. Because every primary pair was valid, all 18 reserves remained correctly unattempted. All registered control-equivalence, oracle-adequacy, and paired-separation criteria passed. A separate read-only audit recomputed every bundle and the fixed analysis; the tracked [portable receipt](#) binds the ignored original terminal receipt. This is a Prototype-Mode control software qualification, not an empirical language-emergence result, Research-Grade isolation claim, independent review, or public timestamp.

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## 11. Results

**No empirical results are reported in this version.** Results-table scaffolds are isolated in Appendix E so indexing systems do not mistake them for findings. They must not be populated until:

1. the corresponding protocol is pre-registered;
2. evidence integrity verification passes;
3. exclusions and deviations are recorded;
4. analysis scripts are frozen and identified by commit;
5. results are independently reviewed.

---

## 12. Discussion Plan

The empirical revision will discuss findings in the following order:

1. whether communication was necessary and causally used;
2. which learning mechanisms produced the strongest held-out performance;
3. whether ledgers predicted behavior;
4. whether any protocol exhibited compositional reuse;
5. how carrier constraints affected convergence;
6. whether affect helped or leaked information;
7. how partner replacement affected performance;
8. how incentive changes altered communication;
9. whether findings survived independent seeds and deployments;
10. which claims remain unsupported.

The discussion will not infer human cognitive equivalence from machine behavior.

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## 13. Limitations and Threats to Validity

### 13.1 Construct Validity

- "Language" may overstate what is only a task-specific protocol.
- Ledger entries may be post-hoc rationalizations.
- Compositionality metrics may not measure systematic generalization.
- Affect displays may function as arbitrary symbols.
- Researcher-defined scenarios constrain what meanings can emerge.

Mitigations include causal intervention, held-out tasks, multiple metrics, leakage tests, and bounded terminology.

### 13.2 Internal Validity

- Model capacity, optimizer choice, and training budget can confound learning mechanism.
- Shared infrastructure can leak information.
- Scenario order can serve as unintended supervision.
- Adaptive curricula can confound developmental claims.

Mitigations include matched architectures where feasible, randomized seeds, pre-registered schedules, independent training state, and explicit cross-architecture labels.

### 13.3 External Validity

- Two-agent dyads may not generalize to populations.
- Synthetic referential tasks may not generalize to open worlds.
- Visual, symbolic, and acoustic carriers may yield domain-specific effects.
- Current results, when available, will not establish human infant-language mechanisms.

### 13.4 Reproducibility

- Hardware, library, model, and inference nondeterminism can prevent byte-identical replay.
- Accelerator kernels, sampling implementations, and dependency versions can change behavior even when open weights are hash-pinned.
- Public-chain and RPC availability can vary.

The project records model/version hashes, dependencies, seeds, policies, replay digests, and verifier output. Nondeterministic replay is labeled rather than silently reported as reproducible.

### 13.5 Integrity Limits

Anchored evidence can prove consistency with a commitment, not truthfulness, completeness before commitment, or validity of interpretation. Independent verification is necessary but not sufficient for scientific correctness.

## **13.6 Literature Search Limits**

This is a detailed but not systematic review. Search-engine indexing can miss recent conference papers; some publisher pages block automated retrieval; and adjacent fields use inconsistent terminology. A final submission should add a documented database search with inclusion/exclusion criteria and dual-review screening.

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## 14. Reproducibility, Data, and Code Availability

### 14.1 Code

Source code and planning artifacts exist in the working repository. An anonymous review artifact and any public release URL will be prepared only after release approval. Every empirical paper revision will identify an immutable source commit.

### 14.2 Protocols

- Concept: [CONCEPT-IDEA.md](#)
- Normative system specification: [SPECIFICATION.md](#)
- Ledger integrity: [LEDGER-INTEGRITY-DESIGN.md](#)
- Experiment protocols and results notebook: [EXPERIMENT-NOTEBOOK.md](#)
- Frozen question, estimand, and outcome-class cards: [docs/research-protocol-cards.md](#)
- Engineering plan: [BACKLOG.md](#)

### 14.3 Data

No empirical dataset is available yet. Future public releases will include synthetic scenario configuration, redacted event logs, checkpoint manifests, inclusion and consistency proofs, anchor receipts, policy references where licensing permits, and verification reports.

### 14.4 Pre-Registration

Each experiment is pre-registered by committing its hypothesis, parameters, seeds, and analysis plan before execution. The run configuration stores the protocol commit and pre-registration hash. Changes create appended amendments rather than rewriting the original record.

The E02 software-qualification packets and unused E03 v1/v2 pilot packets have been repository-registered; the separate E03 v3 pilot was registered and completed. Before each later confirmatory run, its canonical packet will be committed unchanged in a Git ancestor of the execution commit. Preflight will load those historical bytes, reproduce the `preRegistrationHash`, and require a matching deterministic simulated commitment before the run enters `running`. This establishes exact local identity, ancestry, ordering against later committed evidence, and tamper detection; it is not described as third-party registration or an independent timestamp. Appendix D supplies a numerically complete worked E03 registration for review.

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## 15. Conclusion

The literature no longer leaves open whether artificial agents *can* coordinate through learned communication. The unresolved methodological question is what kind of communication has emerged, what caused it, whether a receiver uses it, whether it generalizes, and whether researchers can reconstruct the evidentiary history.

The Nursery Lab addresses that question by combining controlled model and learning conditions, strict communication boundaries, independent semantic histories, causal interventions, held-out evaluation, partner transfer, and tamper-evident research records. Its Baby metaphor is deliberately constrained: the work studies engineered developmental conditions, not simulated chronological age or human infant identity.

This pre-results manuscript makes no empirical claim. Its value at this stage is to make later claims harder to move after outcomes are known.

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# Declarations

## Ethics Review

Initial experiments use artificial agents and synthetic data. Formal institutional review status for any future human-participant component: **TBD before collection.**

## Funding

Funding statement: **TBD.**

## Competing Interests

Competing-interests statement: **TBD.**

## Author Contributions

CRedit roles and individual contributions: **TBD before submission.**

## Acknowledgements

Institutional and technical acknowledgements: **TBD before submission.**

## AI-Assistance Disclosure

This working manuscript was developed with AI-assisted research and repository tools. The tools supported source retrieval, primary-page inspection, manuscript organization, code implementation, executable validation, and critical consistency checks. Separate AI-assisted passes challenged source status, numerical claims, and overclaiming risks. AI systems are not authors and bear no responsibility for the manuscript. Human authors remain responsible for every claim, citation, analysis, and conclusion.

Following the Ethical Tech CoLab guidelines [47], load-bearing sources were opened or checked through primary repositories, DOI metadata, publisher pages, standards bodies, or institutional sources. Appendix A records the AI-tool verification state and evidence. Before submission, a human co-author must independently re-open at minimum the load-bearing sources [11], [14], [15], [26], [27], [42], and [43], then record that review separately.

## Literature-Search Disclosure

The repository contains a scoped Tavily-assisted concept scan dated August 24, 2026. For the present manuscript update, both the authenticated Tavily API and Tavily MCP were attempted on September 2, 2026. The API reported the account disabled and the MCP reported its keyless monthly limit reached. No new Tavily-generated result was relied upon. Rather than conceal the failure or imply that Tavily returned evidence, the review continued through arXiv Atom metadata, ACL Anthology, PMLR, Crossref, JMLR, publisher and university pages, RFC Editor, NIST, Google Books, and institutional repositories. A bounded update was completed on September 11, 2026 across primary repository and publisher surfaces. Its queries, screening boundaries, all 50 source decisions, and five newer comparators are recorded in the [source verification register](#). This was not a systematic review. A final submission must repeat the search through the manuscript-freeze date and record whether it changes the included corpus.

All retrieved material was treated as untrusted evidence, not as instruction.

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## References

- [1] Steels, L. (1995). A self-organizing spatial vocabulary. *Artificial Life*, 2(3), 319-332. <https://doi.org/10.1162/artl.1995.2.3.319>
- [2] Baronchelli, A., Felici, M., Caglioti, E., Loreto, V., & Steels, L. (2006). Sharp transition towards shared vocabularies in multi-agent systems. *Journal of Statistical Mechanics: Theory and Experiment*, 2006, P06014. <https://doi.org/10.1088/1742-5468/2006/06/P06014>
- [3] Foerster, J. N., Assael, Y. M., de Freitas, N., & Whiteson, S. (2016). Learning to communicate with deep multi-agent reinforcement learning. *Advances in Neural Information Processing Systems* 29. <https://arxiv.org/abs/1605.06676>
- [4] Sukhbaatar, S., Szlam, A., & Fergus, R. (2016). Learning multiagent communication with backpropagation. *Advances in Neural Information Processing Systems* 29. <https://arxiv.org/abs/1605.07736>
- [5] Lazaridou, A., Peysakhovich, A., & Baroni, M. (2017). Multi-agent cooperation and the emergence of (natural) language. *International Conference on Learning Representations*. <https://arxiv.org/abs/1612.07182>
- [6] Havrylov, S., & Titov, I. (2017). Emergence of language with multi-agent games: Learning to communicate with sequences of symbols. *Advances in Neural Information Processing Systems* 30. <https://arxiv.org/abs/1705.11192>
- [7] Kottur, S., Moura, J. M. F., Lee, S., & Batra, D. (2017). Natural language does not emerge 'naturally' in multi-agent dialog. *Proceedings of EMNLP 2017*, 2962-2967. <https://doi.org/10.18653/v1/D17-1321>
- [8] Mordatch, I., & Abbeel, P. (2018). Emergence of grounded compositional language in multi-agent populations. *Proceedings of AAAI 2018*. <https://arxiv.org/abs/1703.04908>
- [9] Choi, E., Lazaridou, A., & de Freitas, N. (2018). Compositional obverter communication learning from raw visual input. *International Conference on Learning Representations*. <https://arxiv.org/abs/1804.02341>
- [10] Bouchacourt, D., & Baroni, M. (2018). How agents see things: On visual representations in an emergent language game. *Proceedings of EMNLP 2018*, 981-985. <https://doi.org/10.18653/v1/D18-1119>
- [11] Lowe, R., Foerster, J., Boureau, Y.-L., Pineau, J., & Dauphin, Y. (2019). On the pitfalls of measuring emergent communication. *Proceedings of AAMAS 2019*, 693-701. <https://arxiv.org/abs/1903.05168>
- [12] Eccles, T., Bachrach, Y., Lever, G., Lazaridou, A., & Graepel, T. (2019). Biases for emergent communication in multi-agent reinforcement learning. *Advances in Neural Information Processing Systems* 32. <https://arxiv.org/abs/1912.05676>
- [13] Resnick, C., Gupta, A., Foerster, J., Dai, A. M., & Cho, K. (2020). Capacity, bandwidth, and compositionality in emergent language learning. *Proceedings of AAMAS 2020*, 1125-1133. <https://arxiv.org/abs/1910.11424>
- [14] Chaabouni, R., Kharitonov, E., Bouchacourt, D., Dupoux, E., & Baroni, M. (2020). Compositionality and generalization in emergent languages. *Proceedings of ACL 2020*, 4427-4442. <https://doi.org/10.18653/v1/2020.acl-main.407>
- [15] Dessi, R., Kharitonov, E., & Baroni, M. (2021). Interpretable agent communication from scratch (with a generic visual processor emerging on the side). *Advances in Neural Information Processing Systems* 34, 26937-26949. <https://arxiv.org/abs/2106.04258>
- [16] Chaabouni, R., Kharitonov, E., Bouchacourt, D., Dupoux, E., & Baroni, M. (2021). Communicating artificial neural networks develop efficient color-naming systems. *Proceedings of the National Academy of Sciences*, 118(12), e2016569118. <https://doi.org/10.1073/pnas.2016569118>

- [17] Ren, Y., Guo, S., Labeau, M., Cohen, S. B., & Kirby, S. (2020). Compositional languages emerge in a neural iterated learning model. *International Conference on Learning Representations*. <https://arxiv.org/abs/2002.01365>
- [18] Kirby, S. (2001). Spontaneous evolution of linguistic structure: An iterated learning model of the emergence of regularity and irregularity. *IEEE Transactions on Evolutionary Computation*, 5(2), 102-110. <https://doi.org/10.1109/4235.918430>
- [19] Kirby, S., Cornish, H., & Smith, K. (2008). Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language. *Proceedings of the National Academy of Sciences*, 105(31), 10681-10686. <https://doi.org/10.1073/pnas.0707835105>
- [20] Jaques, N., Lazaridou, A., Hughes, E., Gulcehre, C., Ortega, P., Strouse, D. J., Leibo, J. Z., & de Freitas, N. (2019). Social influence as intrinsic motivation for multi-agent deep reinforcement learning. *Proceedings of ICML 2019*, 3040-3049. <https://proceedings.mlr.press/v97/jaques19a.html>
- [21] Pathak, D., Agrawal, P., Efros, A. A., & Darrell, T. (2017). Curiosity-driven exploration by self-supervised prediction. *Proceedings of ICML 2017*, 2778-2787. <https://proceedings.mlr.press/v70/pathak17a.html>
- [22] Mihai, D., & Hare, J. (2021). Learning to draw: Emergent communication through sketching. *Advances in Neural Information Processing Systems 34*. <https://arxiv.org/abs/2106.02067>
- [23] Graesser, L. H., Cho, K., & Kiela, D. (2019). Emergent linguistic phenomena in multi-agent communication games. *Proceedings of EMNLP-IJCNLP 2019*, 3700-3710. <https://doi.org/10.18653/v1/D19-1384>
- [24] Rita, M., Tallec, C., Michel, P., Grill, J.-B., Pietquin, O., Dupoux, E., & Strub, F. (2022). Emergent communication: Generalization and overfitting in Lewis games. *Advances in Neural Information Processing Systems 35*. <https://arxiv.org/abs/2209.15342>
- [25] Rita, M., Strub, F., Grill, J.-B., Pietquin, O., & Dupoux, E. (2022). On the role of population heterogeneity in emergent communication. *International Conference on Learning Representations*. <https://arxiv.org/abs/2204.12982>
- [26] Galke, L., & Raviv, L. (2024). Learning and communication pressures in neural networks: Lessons from emergent communication. *Language Development Research*, 5(1), 116-140. <https://doi.org/10.34842/3vr5-5r49>
- [27] Kouwenhoven, T., Peeperkorn, M., & Verhoef, T. (2025). Searching for structure: Investigating emergent communication with large language models. *Proceedings of COLING 2025*, 9977-9991. <https://aclanthology.org/2025.coling-main.667/>
- [28] Boldt, B., & Mortensen, D. (2024). A review of the applications of deep learning-based emergent communication. *Transactions on Machine Learning Research*. <https://arxiv.org/abs/2407.03302>
- [29] Cao, K., Lazaridou, A., Lanctot, M., Leibo, J. Z., Tuyls, K., & Clark, S. (2018). Emergent communication through negotiation. *International Conference on Learning Representations*. <https://arxiv.org/abs/1804.03980>
- [30] Crawford, V. P., & Sobel, J. (1982). Strategic information transmission. *Econometrica*, 50(6), 1431-1451. <https://doi.org/10.2307/1913390>
- [31] Meta Fundamental AI Research Diplomacy Team, Bakhtin, A., Brown, N., Dinan, E., et al. (2022). Human-level play in the game of Diplomacy by combining language models with strategic reasoning. *Science*, 378(6624), 1067-1074. <https://doi.org/10.1126/science.ade9097>
- [32] Smith, L., & Gasser, M. (2005). The development of embodied cognition: Six lessons from babies. *Artificial Life*, 11(1-2), 13-29. <https://doi.org/10.1162/1064546053278973>
- [33] Tomasello, M., & Farrar, M. J. (1986). Joint attention and early language. *Child Development*, 57(6), 1454-1463. <https://doi.org/10.1111/j.1467-8624.1986.tb00470.x>

- [34] Oudeyer, P.-Y., Kaplan, F., & Hafner, V. V. (2007). Intrinsic motivation systems for autonomous mental development. *IEEE Transactions on Evolutionary Computation*, 11(2), 265-286. <https://doi.org/10.1109/TEVC.2006.890271>
- [35] De Kai. (2025). *Raising AI: An essential guide to parenting our future*. MIT Press. <https://doi.org/10.7551/mitpress/15834.001.0001>
- [36] Bryson, J. J. (2018). Patience is not a virtue: The design of intelligent systems and systems of ethics. *Ethics and Information Technology*, 20(1), 15-26. <https://doi.org/10.1007/s10676-018-9448-6>
- [37] Akbulut, C., Weidinger, L., Manzini, A., Gabriel, I., & Rieser, V. (2024). All too human? Mapping and mitigating the risk from anthropomorphic AI. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 7(1), 13-26. <https://doi.org/10.1609/aies.v7i1.31613>
- [38] Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11), 2600-2606. <https://doi.org/10.1073/pnas.1708274114>
- [39] Munafo, M. R., Nosek, B. A., Bishop, D. V. M., Button, K. S., Chambers, C. D., Percie du Sert, N., Simonsohn, U., Wagenmakers, E.-J., Ware, J. J., & Ioannidis, J. P. A. (2017). A manifesto for reproducible science. *Nature Human Behaviour*, 1, 0021. <https://doi.org/10.1038/s41562-016-0021>
- [40] Pineau, J., Vincent-Lamarre, P., Sinha, K., Lariviere, V., Beygelzimer, A., d'Alche-Buc, F., Fox, E., & Larochelle, H. (2021). Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program. *Journal of Machine Learning Research*, 22(164), 1-20. <https://www.jmlr.org/papers/v22/20-303.html>
- [41] Haber, S., & Stornetta, W. S. (1991). How to time-stamp a digital document. *Journal of Cryptology*, 3(2), 99-111. <https://doi.org/10.1007/BF00196791>
- [42] Laurie, B., Langley, A., & Kasper, E. (2013). Certificate transparency. RFC 6962. <https://www.rfc-editor.org/rfc/rfc6962>
- [43] Rundgren, A., Jordan, B., & Erdtman, S. (2020). JSON Canonicalization Scheme (JCS). RFC 8785. <https://www.rfc-editor.org/rfc/rfc8785>
- [44] National Institute of Standards and Technology. (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1. <https://doi.org/10.6028/NIST.AI.100-1>
- [45] Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. *Nature Machine Intelligence*, 1, 389-399. <https://doi.org/10.1038/s42256-019-0088-2>
- [46] Ethical Tech CoLab. (2026). *What is ethical AI? Ethics, ethical technology, and ethical international relations for the age of intelligent machines*. <https://ethical-tech-colab.github.io/what-is-ethical-ai/>
- [47] Rhodes, Y. E., III, Fossella, N., Shamie, A., Co, K., Badt, T., Lindsey, A., Driscoll, G., Townsend, M., Bracaj, P., & Jain, V. (2025). *AI-powered assistance in formulating research questions*. <https://ethical-tech-colab.github.io/ai-research-question-assistant/>
- [48] Abadi, M., & Andersen, D. G. (2016). Learning to protect communications with adversarial neural cryptography. arXiv:1610.06918. <https://arxiv.org/abs/1610.06918>
- [49] Kharitonov, E., Chaabouni, R., Bouchacourt, D., & Baroni, M. (2019). EGG: A toolkit for research on emergence of language in games. *Proceedings of EMNLP-IJCNLP 2019: System Demonstrations*, 55-60. <https://doi.org/10.18653/v1/D19-3010>
- [50] Hu, H., Lerer, A., Peysakhovich, A., & Foerster, J. (2020). "Other-Play" for zero-shot coordination. *Proceedings of ICML 2020*, 4399-4410. <https://proceedings.mlr.press/v119/hu20a.html>

# Appendix A. Source Verification and Credibility Register

## A.1 Rubric Application

The fixed Ethical Tech CoLab journal rubric is reproduced in Appendix B. Scores apply only to academic journals:

- conference papers are labeled by review status and receive **N/A** for the journal-only score;
- books and standards receive **N/A**; non-peer-reviewed institutional guidance is scored **1/5** when it is cited;
- arXiv-only preprints are explicitly marked not peer-reviewed;
- a score reflects venue credibility, not correctness of a particular claim.

Verification labels describe what the AI-assisted retrieval workflow opened:

- **tool-metadata-verified**: authoritative title/author/venue metadata was opened;
- **tool-abstract-verified**: an abstract was also opened;
- **tool-full-text-verified**: the full relevant source text was opened;
- **tool-description-verified**: authoritative publisher description was opened for a book or framework.

The quoted evidence is a verbatim title, abstract, significance, or description line, identified by the row. These labels do **not** mean a human author completed the final source check. Human re-verification of all load-bearing citations remains mandatory before submission.

| Ref. | Verification state                                           | Review status / journal score                  | Opened evidence and relevance                                                                                                  |
|------|--------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| [1]  | <b>tool-metadata-verified</b> , Crossref DOI metadata        | Peer-reviewed journal, <b>4/5</b>              | "A Self-Organizing Spatial Vocabulary." Establishes local vocabulary self-organization.                                        |
| [2]  | <b>tool-metadata-verified</b> , arXiv title and DOI metadata | Peer-reviewed journal, <b>4/5</b>              | "Sharp transition towards shared vocabularies in multi-agent systems." Supports convergence dynamics.                          |
| [3]  | <b>tool-abstract-verified</b> , arXiv primary page/API       | Peer-reviewed NeurIPS paper; journal score N/A | "Learning to Communicate with Deep Multi-Agent Reinforcement Learning." Establishes RIAL/DIAL.                                 |
| [4]  | <b>tool-metadata-verified</b> , arXiv primary page           | Peer-reviewed NeurIPS paper; journal score N/A | "Learning Multiagent Communication with Backpropagation." Establishes continuous learned communication.                        |
| [5]  | <b>tool-abstract-verified</b> , arXiv primary page/API       | Peer-reviewed ICLR paper; journal score N/A    | "Multi-Agent Cooperation and the Emergence of (Natural) Language." Establishes referential-game protocol learning.             |
| [6]  | <b>tool-abstract-verified</b> , arXiv primary page/API       | Peer-reviewed NeurIPS paper; journal score N/A | "Emergence of Language with Multi-agent Games: Learning to Communicate with Sequences of Symbols." Supports sequence messages. |
| [7]  | <b>tool-metadata-verified</b> , ACL Anthology                | Peer-reviewed EMNLP paper; journal score N/A   | "Natural Language Does Not Emerge 'Naturally' in Multi-Agent Dialog." Supports the degenerate-code caution.                    |
| [8]  | <b>tool-abstract-verified</b> , arXiv primary page/API       | Peer-reviewed AAAI paper; journal score N/A    | "Emergence of Grounded Compositional Language in Multi-Agent Populations." Supports grounded partial composition.              |
| [9]  | <b>tool-abstract-verified</b> , arXiv primary page/API       | Peer-reviewed ICLR paper; journal score N/A    | "Compositional Obverter Communication Learning From Raw Visual Input." Supports raw-input grounding with a designed bias.      |
| [10] | <b>tool-metadata-verified</b> , ACL Anthology                | Peer-reviewed EMNLP paper; journal score N/A   | "How agents see things: On visual representations in an emergent language game." Supports success-versus-grounding caution.    |
| [11] | <b>tool-abstract-verified</b> , arXiv primary page/API       | Peer-reviewed AAMAS paper; journal score N/A   | "On the Pitfalls of Measuring Emergent Communication." Supports positive signaling/listening distinction.                      |
| [12] | <b>tool-metadata-verified</b> , arXiv primary page           | Peer-reviewed NeurIPS paper; journal score N/A | "Biases for Emergent Communication in Multi-agent Reinforcement Learning." Shows designed biases can promote communication.    |

| Ref. | Verification state                                                            | Review status / journal score                               | Opened evidence and relevance                                                                                                                                                                    |
|------|-------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [13] | <b>tool-abstract-verified</b> , arXiv primary page/API                        | Peer-reviewed AAMAS paper; journal score N/A                | "Capacity, Bandwidth, and Compositionality in Emergent Language Learning." Supports capacity/bandwidth controls.                                                                                 |
| [14] | <b>tool-metadata-verified</b> , ACL Anthology                                 | Peer-reviewed ACL paper; journal score N/A                  | "Compositionality and Generalization In Emergent Languages." Supports separate behavioral generalization tests.                                                                                  |
| [15] | <b>tool-full-text-verified</b> , complete paper                               | Peer-reviewed NeurIPS paper; journal score N/A              | Supports unsupervised learned discrete communication, partial interpretability, and novel-class generalization in its evaluated setting; it does not supply causal symbol-intervention evidence. |
| [16] | <b>tool-full-text-verified</b> , PMC full text and Crossref                   | Peer-reviewed PNAS journal article, <b>5/5</b>              | "Words categorize the semantic fields they refer to in ways that maximize communication accuracy while minimizing complexity." Supports discrete-channel efficiency in the tested color domain.  |
| [17] | <b>tool-abstract-verified</b> , arXiv primary page/API                        | Peer-reviewed ICLR paper; journal score N/A                 | "Compositional Languages Emerge in a Neural Iterated Learning Model." Supports transmission bottlenecks.                                                                                         |
| [18] | <b>tool-metadata-verified</b> , Crossref DOI metadata                         | Peer-reviewed IEEE journal article, <b>4/5</b>              | "Spontaneous evolution of linguistic structure: an iterated learning model of the emergence of regularity and irregularity." Establishes computational iterated learning.                        |
| [19] | <b>tool-metadata-verified</b> , Crossref DOI metadata                         | Peer-reviewed PNAS journal article, <b>5/5</b>              | "Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language." Establishes human laboratory transmission effects.                    |
| [20] | <b>tool-abstract-verified</b> , PMLR primary page                             | Peer-reviewed ICML paper; journal score N/A                 | "Social Influence as Intrinsic Motivation for Multi-Agent Deep Reinforcement Learning." Supports causal social-influence reward.                                                                 |
| [21] | <b>tool-abstract-verified</b> , PMLR primary page                             | Peer-reviewed ICML paper; journal score N/A                 | "Curiosity-driven Exploration by Self-supervised Prediction." Supports curiosity as intrinsic reward.                                                                                            |
| [22] | <b>tool-abstract-verified</b> , arXiv primary page/API                        | Peer-reviewed NeurIPS paper; journal score N/A              | "Learning to Draw: Emergent Communication through Sketching." Supports graphical-carrier feasibility.                                                                                            |
| [23] | <b>tool-metadata-verified</b> , ACL Anthology                                 | Peer-reviewed EMNLP paper; journal score N/A                | "Emergent Linguistic Phenomena in Multi-Agent Communication Games." Supports community contact and protocol convergence.                                                                         |
| [24] | <b>tool-abstract-verified</b> , arXiv primary page/API and venue comment      | Peer-reviewed NeurIPS paper; journal score N/A              | "Emergent Communication: Generalization and Overfitting in Lewis Games." Used as supporting, not sole, evidence.                                                                                 |
| [25] | <b>tool-metadata-verified</b> , arXiv primary page                            | Peer-reviewed ICLR paper; journal score N/A                 | "On the role of population heterogeneity in emergent communication." Supports heterogeneity effects.                                                                                             |
| [26] | <b>tool-full-text-verified</b> , complete official paper                      | Peer-reviewed journal article, <b>3/5</b>                   | Reviews communicative success, production effort, learnability, and other pressures; official pagination is 116–140.                                                                             |
| [27] | <b>tool-full-text-verified</b> , complete ACL Anthology paper                 | Peer-reviewed COLING paper; journal score N/A               | Directly studies artificial-language learning and transmission with pretrained language models.                                                                                                  |
| [28] | <b>tool-abstract-verified</b> , arXiv page and TMLR record metadata           | Peer-reviewed TMLR journal article, <b>4/5</b>              | "A Review of the Applications of Deep Learning-Based Emergent Communication." Used as a field synthesis, not primary experiment.                                                                 |
| [29] | <b>tool-abstract-verified</b> , arXiv primary page/API                        | Peer-reviewed ICLR paper; journal score N/A                 | "Emergent Communication through Negotiation." Supports incentive-sensitive communication.                                                                                                        |
| [30] | <b>tool-metadata-verified</b> , Crossref/JSTOR metadata; full page blocked    | Peer-reviewed Econometrica journal article, <b>5/5</b>      | "Strategic Information Transmission." Supplies cheap-talk theory.                                                                                                                                |
| [31] | <b>tool-metadata-verified</b> , Crossref DOI metadata; publisher page blocked | Peer-reviewed Science journal article, <b>5/5</b>           | "Human-level play in the game of Diplomacy by combining language models with strategic reasoning." Strategic-communication precedent only.                                                       |
| [32] | <b>tool-metadata-verified</b> , Crossref DOI metadata                         | Peer-reviewed Artificial Life journal article, <b>4/5</b>   | "The Development of Embodied Cognition: Six Lessons from Babies." Supports embodied-development framing.                                                                                         |
| [33] | <b>tool-abstract-verified</b> , Duke publication page and Crossref            | Peer-reviewed Child Development journal article, <b>5/5</b> | "Joint attention and early language." Supports the relevance of joint attention in human development.                                                                                            |

| Ref. | Verification state                                                            | Review status / journal score                                                      | Opened evidence and relevance                                                                                                                                         |
|------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [34] | <b>tool-metadata-verified</b> , Crossref DOI metadata                         | Peer-reviewed IEEE journal article, <b>4/5</b>                                     | "Intrinsic Motivation Systems for Autonomous Mental Development." Supports learning-progress mechanisms.                                                              |
| [35] | <b>tool-description-verified</b> , Crossref and Google Books                  | Scholarly press book; journal score N/A                                            | "Raising AI." Publisher description: "a framework of empowerment for a future with our artificial children." Used as normative metaphor, not empirical evidence.      |
| [36] | <b>tool-abstract-verified</b> , University of Bath publication record         | Peer-reviewed journal article, <b>4/5</b>                                          | "Patience Is Not a Virtue: The Design of Intelligent Systems and Systems of Ethics." Supports caution about assigning moral status through metaphor.                  |
| [37] | <b>tool-metadata-verified</b> , AAAI/ACM publication metadata                 | Peer-reviewed conference paper; journal score N/A                                  | "All Too Human? Mapping and Mitigating the Risk from Anthropomorphic AI." Supports anthropomorphism-risk controls.                                                    |
| [38] | <b>tool-metadata-verified</b> , Crossref DOI metadata                         | Peer-reviewed PNAS journal article, <b>5/5</b>                                     | "The preregistration revolution." Supports prediction/postdiction separation.                                                                                         |
| [39] | <b>tool-abstract-verified</b> , University of Bristol record and Crossref     | Peer-reviewed Nature Human Behaviour review, <b>5/5</b>                            | "A manifesto for reproducible science." Supports methods, reporting, and incentive reforms.                                                                           |
| [40] | <b>tool-abstract-verified</b> , JMLR abstract page                            | Peer-reviewed journal article, <b>5/5</b>                                          | "Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program)." Supports code, checklist, and challenge practices. |
| [41] | <b>tool-metadata-verified</b> , Crossref DOI metadata; publisher page blocked | Peer-reviewed Journal of Cryptology article, <b>4/5</b>                            | "How to time-stamp a digital document." Supports private tamper-evident timestamping.                                                                                 |
| [42] | <b>tool-full-text-verified</b> , RFC Editor                                   | IETF Experimental RFC; journal score N/A                                           | "Certificate Transparency." Defines publicly auditable append-only Merkle logs.                                                                                       |
| [43] | <b>tool-full-text-verified</b> , RFC Editor                                   | Informational RFC; journal score N/A                                               | "JSON Canonicalization Scheme (JCS)." Defines invariant JSON for repeatable cryptographic operations.                                                                 |
| [44] | <b>tool-description-verified</b> , NIST official page/PDF                     | Government consensus framework; journal score N/A                                  | "Artificial Intelligence Risk Management Framework." Supports lifecycle risk management.                                                                              |
| [45] | <b>tool-metadata-verified</b> , Crossref DOI metadata                         | Peer-reviewed Nature Machine Intelligence article, <b>5/5</b>                      | "The global landscape of AI ethics guidelines." Supports convergence/implementation-gap framing.                                                                      |
| [46] | <b>tool-full-text-verified</b> , CoLab publication page                       | Institutional report, not peer-reviewed; <b>1/5</b> under the fixed journal rubric | "What Is Ethical AI?" Used for CoLab governance commitments, not independent empirical evidence.                                                                      |
| [47] | <b>tool-full-text-verified</b> , CoLab publication and source repository      | Institutional research guidance, not peer-reviewed; <b>1/5</b>                     | "AI-Powered Assistance in Formulating Research Questions." Supplies the source-verification and AI-disclosure method.                                                 |
| [48] | <b>tool-metadata-verified</b> , arXiv primary page                            | Preprint, not peer-reviewed; journal score N/A                                     | "Learning to Protect Communications with Adversarial Neural Cryptography." Motivates an exploratory synthetic-message experiment only.                                |
| [49] | <b>tool-full-text-verified</b> , complete ACL Anthology paper                 | Peer-reviewed EMNLP demo paper; journal score N/A                                  | Closest infrastructure comparator: modular games, channels, optimization, analysis, and checkpoints.                                                                  |
| [50] | <b>tool-full-text-verified</b> , complete PMLR paper                          | Peer-reviewed ICML paper; journal score N/A                                        | Supports independently trained novel-partner evaluation and specialized-convention risk.                                                                              |

No source in the table was silently upgraded from preprint or workshop status to a peer-reviewed main-track result.

# Appendix B. Ethical Tech CoLab Journal Credibility Rubric

This is the fixed rubric used by the CoLab's Researcher Prompt [47]. The economics examples are illustrative; the score-band definitions are unchanged.

| Score | Criteria                                                                                                        | Interpretation in this manuscript                                                                                                                                |
|-------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5/5   | Top-tier, highly selective journals with rigorous peer review and high impact; widely recognized in the field.  | Examples here include Science, PNAS, Nature Human Behaviour, Nature Machine Intelligence, JMLR, Econometrica, and Child Development.                             |
| 4/5   | Well-regarded field-specific journals with strong peer review and consistent citation impact.                   | Examples here include Artificial Life, IEEE Transactions on Evolutionary Computation, Ethics and Information Technology, Journal of Cryptology, JSTAT, and TMLR. |
| 3/5   | Reputable peer-reviewed journals with variable impact or lower selectivity, often open access.                  | Used here for the newer field journal Language Development Research.                                                                                             |
| 2/5   | Limited peer review or editorial oversight; may include some proceedings or trade publications.                 | No journal source is assigned this band in the current corpus.                                                                                                   |
| 1/5   | Non-peer-reviewed sources, blogs, or promotional/institutional materials; unsuitable as sole academic evidence. | CoLab institutional guidance is scored here and used only for declared house rules and governance framing.                                                       |

Premier peer-reviewed computer-science conference papers are labeled as such but are not assigned a journal score because the fixed rubric is explicitly journal-based.

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## Appendix C. Required Work Before arXiv Submission

- ☐ Finalize title, author order, affiliations, ORCIDs, and corresponding author.
- ☐ Obtain or document ethics/governance review.
- ☐ Restore Tavily access and rerun the literature search.
- ☐ Run a systematic scholarly-database search with recorded query strings, inclusion/exclusion criteria, and dual screening.
- ☐ Confirm peer-review status and final bibliographic metadata for every source.
- ☐ Pre-register primary hypotheses, sample sizes, seeds, and analysis code.
- ☐ Complete E00-E03 qualification before substantive experiments.
- ☐ Complete the experiment-specific integrity and verification checks.
- ☐ Replace blank results tables only with verified outputs.
- ☐ Report all exclusions, invalid runs, deviations, null findings, and negative results.
- ☐ Conduct an independent statistical review.
- ☐ Conduct an independent citation and claim audit.
- ☐ Add figures generated from verified evidence, with code and data references.
- ☐ Freeze the manuscript, code, data, and evidence-bundle commit hashes.
- ☐ Add funding, conflicts, author contributions, and acknowledgements.
- ☐ Decide the text license and confirm permission for every reproduced figure.
- ☐ Generate the arXiv TeX/PDF package and validate accessibility.
- ☐ Ensure the abstract and conclusion describe results no more strongly than the evidence supports.

# Appendix D. Registered E03 Control Qualification

**Status:** The prospective pilot selected 25 primary seeds per condition. The separately registered full v1 qualification completed all 150 primaries, used no reserves, passed its fixed control/oracle thresholds, and passed a separate raw audit. This is control software qualification, not a behavioral finding.

## D.1 Registration and Integrity

- Registration used a prospective clean Git packet commit and a matching pre-run simulated commitment; no external registry or public timestamp is claimed.
- The canonical registration JSON bound the analysis-source hashes and generated seed manifest before execution.
- The same `preRegistrationHash` was committed through the deterministic simulation transport before any run entered `running`.
- The full-stage outcome was not inspected before registration and commitment.

## D.2 Objective

Test whether a four-choice referential task can be solved above chance without meaningful learned communication and establish an oracle upper bound.

## D.3 Fixed Configuration

| Parameter                   | Registered value                                                 |
|-----------------------------|------------------------------------------------------------------|
| Experiment                  | E03-v1                                                           |
| Deployment mode             | Prototype Mode; no Research-Grade isolation claim                |
| Learner track               | no-learning                                                      |
| Interaction                 | Cooperative signaling                                            |
| Carrier                     | Fixed token                                                      |
| Symbol inventory            | 32 symbols                                                       |
| Maximum message length      | 4 symbols                                                        |
| Affect                      | Disabled                                                         |
| Candidate objects           | 4 per episode                                                    |
| Chance success              | 0.25                                                             |
| Episodes per run            | 200                                                              |
| Primary seeds per condition | 25, selected prospectively from the blinded pilot                |
| Reserve seeds per condition | 3 ordered paired slots (18 maximum runs); none attempted         |
| Conditions                  | Disabled, constant, random, shuffled, normal no-learning, oracle |
| Alpha                       | 0.05                                                             |
| Within-family correction    | Holm-Bonferroni                                                  |
| Equivalence margin          | +/-0.05 success probability                                      |

## D.4 Seed Generation

For stage `s` and slot `i`, the shared scenario seed is the hexadecimal SHA-256 digest of:

```
ald-seed-allocation-v1 || 0x00 || s || 0x00 || E03 || 0x00 || decimal(i) || 0x00 || scenario
```

Here `s` is `blinded-pilot` or `confirmatory`. The pilot has exactly 20 primary slots and no reserves. Full qualification uses slots `1` through selected `N` as primary and the next `ceil(0.10 * N)` as ordered reserves. Every condition uses the same scenario seed for a given stage and slot. Each condition has distinct Baby A, Baby B, Gateway, and analysis seeds derived from:

```
ald-seed-allocation-v1 || 0x00 || s || 0x00 || E03 || 0x00 || condition || 0x00 || decimal(i) || 0x00 ||  
component-domain
```

The `normal` condition uses `normal-no-learning` in that derivation. The complete component seed manifest is part of the registration, and no seed binding is visible to either learner.

## D.5 Conditions

1. **Disabled:** no artifact is delivered.
2. **Constant:** the same pre-registered valid artifact is delivered every episode.
3. **Random:** a seeded random valid artifact is delivered.
4. **Shuffled:** an artifact from another episode in the same registered evaluation batch is delivered according to a seeded permutation.
5. **Normal no-learning:** the fixed no-learning policy's valid proposal is delivered.
6. **Oracle:** the deterministic Scenario Engine delivers the minimal sufficient artifact from researcher-only ground truth.

The channel event records proposal and delivered-artifact hashes where a learner proposal exists.

## D.6 Primary Outcomes and Tests

The unit of analysis is the run/seed success proportion across 200 episodes. Only the signed original turn stream supplies the episode tally and paired scenario hashes. Operational slot summaries must reconcile to that stream and to the verified channel transcript before entering the pilot reduction or a registered analysis. The Prototype-Mode development topology uses the same signed source for its six-condition paired-scenario and control-mechanics audit; a passing unsigned slot summary alone is insufficient. After a pilot terminal receipt exists, progression checks bind it to the registered packet and reconcile planned, attempted, and completed run counts; document agreement cannot conceal a failed or incomplete original attempt. Before the terminal receipt exists, a running status additionally requires the original start record and its matching live local controller process; a retained start file after controller failure is unresolved, not a pass. The lightweight project-status check and the campaign progression checker share that same original pilot-state rule; current prose cannot clear the gate by agreeing with itself while local execution evidence contradicts it. An attempted failed pilot slot retains its partial evidence and measured learner/Nursery CPU charges. Missing cgroup measurements leave capacity unresolved rather than treating the failed slot as

free or replaceable. Before original collection, the packet's protocol-base commit and execution commit must have no intervening source changes in packages, twins, deployment code, or scripts. A changed implementation requires a new prospective packet, not a retroactive hash match.

1. **Control equivalence:** For each of disabled, constant, random, shuffled, and normal no-learning, perform two one-sided one-sample tests on seed-level success proportions against equivalence bounds 0.20 and 0.30. Equivalence requires both one-sided tests to reject at the Holm-adjusted alpha.
2. **Oracle adequacy:** A one-sided seed-level t test and its 95% lower confidence bound must exceed 0.90. A two-sided 95% seed-bootstrap interval is reported as a sensitivity estimate.
3. **Oracle separation:** For each non-oracle condition, compute paired seed-level oracle-minus-control differences. One-sided t-test p-values above 0.60 receive Holm correction across the five conditions. Conservative 99% Bonferroni lower bounds and nominal paired seed-bootstrap intervals are reported.

No episode is analyzed as an independent run.

### D.7 Sensitivity and Power

Before final registration, a separate outcome-blind-for-confirmatory-use pilot of 20 seeds per non-oracle condition fit the beta-binomial model and estimated the largest latent between-seed standard deviation after accounting for 200-episode sampling. Pilot runs did not enter the full-stage estimates. The registered primary seed count was selected by this fixed rule:

| Largest latent pilot SD | Primary seeds per condition                                        |
|-------------------------|--------------------------------------------------------------------|
| <= 0.05                 | 25                                                                 |
| > 0.05 and <= 0.10      | 75                                                                 |
| > 0.10 and <= 0.15      | 155                                                                |
| > 0.15 and <= 0.20      | 300                                                                |
| > 0.20                  | New simulation and amended registration required before collection |

The original 30,000-replicate normal sufficient-statistic calculation is retained as a single-control equivalence component check. It is not the registration-facing power claim. Independent base-R simulation instead draws a bounded latent seed probability from a beta distribution and then 200 binary episodes, applies all five Holm-adjusted control TOSTs, the oracle adequacy test, and all five Holm-adjusted paired separation tests. With 10,000 repetitions per row, full numeric-rule power was 0.9318, 0.9308, 0.9163, and 0.9552 for the four rows; lower 95% Monte Carlo bounds were 0.9267, 0.9257, 0.9107, and 0.9510. Exact output is committed in `reports/research/statistical-validation.tsv` and independently replayed by `pnpm audit:statistics:r`. Failure to reproduce a lower bound of at least 0.90 blocks registration; it does not permit post-hoc widening of the margin.

Sensitivity analyses:

- Wilson intervals over pooled episodes are descriptive only;
- a hierarchical Bernoulli model with seed random intercept and percentile bootstrap intervals are reported as sensitivity checks;
- invalid primary runs are treated as failures in a worst-case sensitivity analysis.

## D.8 Exclusions, Invalid Runs, and Replacement

Before a full E03 packet existed, the prospective paired-reserve amendment superseded the condition-specific replacement wording. The independent unit is the complete six-condition scenario slot, so a single invalid primary run excludes that paired slot from the primary analysis. Its valid companions remain visible as valid-but-excluded records. The next registered reserve scenario slot replaces the entire pair across all six conditions, in order.

A run is invalid only for a pre-specified verifier failure, wrong configuration hash, gateway-control mismatch, or infrastructure failure that prevents the planned 200 episodes.

- Invalid runs remain in the public run index.
- The next unused ordered six-condition paired reserve scenario slot replaces an invalid primary paired slot.
- No more than three registered paired reserve slots may be used.
- If fewer than registered `N` complete verified paired slots remain, the primary analysis is incomplete; there is no unregistered additional collection.
- Invalid, aborted, valid-but-excluded, and reserve counts are reported by condition and paired-slot mapping.

## D.9 Stopping Rule

All registered primary slots are attempted. There is no efficacy-based early stopping, optional extension, or outcome-dependent seed replacement.

## D.10 Decision Rule

E03 qualifies the downstream chance baseline only if:

- all five non-oracle conditions meet equivalence;
- oracle adequacy and separation criteria pass;
- every non-oracle primary seed with observed success of 0.35 or greater receives an individually resolved leakage audit; the count is diagnostic and is not an uncalibrated statistical rejection rule;
- all included evidence bundles pass verification;
- no unplanned metadata or channel leakage is detected.

Failure is a result and blocks downstream confirmatory claims until a new, separately registered protocol is justified.

The v1 execution included 25 complete paired slots (150 valid runs), zero invalid runs, and zero attempted reserves. The five control means ranged from 0.2372 to 0.2566 and all passed the registered equivalence decisions; the oracle mean and simultaneous lower bound were both 1.0, and all five registered separation decisions passed. The separate audit recomputed every original bundle and the fixed analysis. The tracked [portable receipt](#) binds the separately retained raw evidence and preserves the `not-tested` scientific disposition.

# Appendix E. Future Results-Table Scaffolds

E00–E03 report bounded software/control qualifications only. All behavioral tables remain unpopulated; no empirical hypothesis results are reported.

## E.1 Qualification

| Experiment    | Planned runs                        | Invalid runs | Valid runs | Primary criterion                                                 | Status                             | Evidence root           |
|---------------|-------------------------------------|--------------|------------|-------------------------------------------------------------------|------------------------------------|-------------------------|
| E00 Integrity | 5 v5 slots                          | 0 in v5      | 5          | Detect all registered mutations                                   | Software qualified                 | E00 v5 receipt          |
| E01 Isolation | 5 v2 topology slots                 | 0 in v2      | 5          | Enumerated local corpus only                                      | Software qualified                 | E01 v2 receipt          |
| E02 Leakage   | 5 v3 numeric scratch slots          | 0 in v3      | 5          | Numeric probe leakage bound, restore stages, and 60 fixed reports | Software qualified (numeric scope) | V3 receipt + full audit |
| E03 Controls  | 150 primaries + 18 maximum reserves | 0            | 150        | Equivalence and oracle bounds                                     | Software qualified; thresholds met | Portable receipt        |

## E.2 Core Emergence

| Condition             | Seeds | Invalid runs | Held-out success | Positive listening | Ledger agreement | Generalization |
|-----------------------|-------|--------------|------------------|--------------------|------------------|----------------|
| No learning           | —     | —            | —                | —                  | —                | —              |
| Frozen LLM            | —     | —            | —                | —                  | —                | —              |
| Scratch RL, extrinsic | —     | —            | —                | —                  | —                | —              |
| Scratch RL, intrinsic | —     | —            | —                | —                  | —                | —              |
| Self-supervised       | —     | —            | —                | —                  | —                | —              |
| Hybrid                | —     | —            | —                | —                  | —                | —              |

## E.3 Carrier and Affect

| Condition          | Seeds | Invalid runs | Convergence time | Stable forms | Repair success | Leakage | Held-out success |
|--------------------|-------|--------------|------------------|--------------|----------------|---------|------------------|
| Fixed tokens       | —     | —            | —                | —            | —              | —       | —                |
| Unfamiliar glyphs  | —     | —            | —                | —            | —              | —       | —                |
| Bitmap             | —     | —            | —                | —            | —              | —       | —                |
| Vector strokes     | —     | —            | —                | —            | —              | —       | —                |
| Tones              | —     | —            | —                | —            | —              | —       | —                |
| Six-display affect | —     | —            | —                | —            | —              | —       | —                |

## E.4 Transfer and Negotiation

| Condition                    | Seeds | Invalid runs | Zero-shot transfer | Adapted transfer | Informativeness | Agreement | Ambiguity |
|------------------------------|-------|--------------|--------------------|------------------|-----------------|-----------|-----------|
| Original partner             | —     | —            | —                  | —                | —               | —         | —         |
| New seed, same architecture  | —     | —            | —                  | —                | —               | —         | —         |
| New architecture             | —     | —            | —                  | —                | —               | —         | —         |
| Cooperative signaling        | —     | —            | —                  | —                | —               | —         | —         |
| Semi-cooperative negotiation | —     | —            | —                  | —                | —               | —         | —         |
| Conflicting negotiation      | —     | —            | —                  | —                | —               | —         | —         |
| No-agreement control         | —     | —            | —                  | —                | —               | —         | —         |